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Transitioning From L1 Medium of Instruction to L2 English Medium Instruction: The Role of Translanguaging in Contributing to EFL Students' Language Learning Motivation

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关键词: 語言學習動機 | 跨語言學習經歷 | 英語作為教學媒介 | 二語自我導向

ABSTRACT

Research on English-Medium-Instruction (EMI) highlights inadequate support for students transitioning from L1-medium-instruction to EMI. To date, little research has focused on how L2 students' adaptation to an EMI environment influences their L2 learning motivation (LLM). Translanguaging has shown positive effects on boosting students' LLM, but its advantages for EMI students' LLM have been less explored. Considering the complex and dynamic nature of LLM, this study draws on a reconceptualised framework of the L2 Motivational Self-System in order to examine the interaction between students' translanguaging learning experiences in an EMI context and students' LLM. Specifically, this study will explore (1) how EMI teacher's creation of translanguaging spaces in EMI classroom interactions shapes students' translanguaging learning experience and contributes to students' LLM, and (2) how teachers and students make sense of translanguaging practices in EMI classrooms. This paper argues that the creation of a translanguaging space in EMI classrooms can play a role in enhancing students' LLM and facilitating their transition to a new EMI environment. The findings will have implications for the advancement of translanguaging as a pedagogy, strategies to enhance students' LLM, and language support initiatives as students' transition from L1-Medium-Instruction to L2 EMI contexts.

摘要

關於英語作為教學媒介(EMI)的研究表明,從母語教學過渡到EMI環境的學生缺乏足夠支持。迄今為止,鮮有研究關注二語學生適應EMI環境如何影響其二語學習動機(LLM)。跨語言實踐已被證明對提升學生LLM具有積極作用,但其對EMI學生LLM的促進作用尚未得到充分重視。鑒於LLM的複雜性和動態性,本研究基於重構的二語動機自我系統框架,探討EMI語境下學生的跨語言學習經歷與其LLM之間的互動關係。具體而言,本研究將探究:1)EMI教師在課堂互動中創建的跨語言空間如

[Correction added on October 22, 2025, after first online publication: The Section 4 has been updated in this version.]

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何形塑學生的跨語言學習體驗並促進其LLM;2)教師與學生如何理解EMI課堂中的跨語言實踐。本文認為,在EMI課堂中構建跨語言空間有助於增強學生LLM,並促進其向新EMI環境的過渡。研究結果將為跨語言教學法的發展、動機培養策略的制定,以及學生從母語教學向二語EMI環境過渡期間的語言支持提供重要啟示。

1 | Introduction

English Medium Instruction (EMI) involves teaching academic subjects in English, rather than in the local language, in countries where the majority of the population does not have English as their primary language (Macaro 2018). In Hong Kong (HK), Cantonese is the primary language spoken by the majority of citizens. In primary schools, Chinese-Medium-Instruction is generally used to teach most content subjects, with English taught as a separate subject, to develop students' L1 skills and expose them to English (Poon 2013). At the secondary level, schools offer EMI, Chinese-medium-instruction or mixed medium-of-instruction. Under the fine-tuned policy since 2010, Chinese-medium-instruction schools have been permitted to teach up to 25% of the curriculum in English if students possess the academic capability to learn effectively through EMI (Education Bureau 2009). As a result, around 30% and 40% of schools use EMI for all grades and at least one content subject, respectively (Tai 2022). Hong Kong's distinctive environment offers a valuable setting for examining how context and society influence motivation to learn a second language (L2). This is because English—once the official language during the colonial period and now a global language—hold significant economic importance in HK, leading to a strong preference for EMI among both parents and students (Poon 2013).

Since most primary school classes in HK use Chinese-Medium-Instruction, such a learning environment gives students limited prior experience in learning content subjects through L2 English. These students receive little support for transitioning to EMI secondary schools (Poon 2013) and experience language-related problems in classroom interaction, content understanding, reading learning materials, and writing assignments (Tsou and Kao 2017). An indispensable factor in EMI academic adaptation is students' L2 learning motivation (LLM) (Du and Jackson 2018; Lasagabaster and Doiz 2017). Although motivation plays a determining role in the success of instructed knowledge acquisition and L2 learning (Lamb 2017), its fluctuation and dynamics during students' EMI adjustment have not been sufficiently studied (Jiang et al. 2019).

Translanguaging has been demonstrated to be an effective pedagogical practice that facilitates content and language learning in EMI settings (Tai and Li 2020; Tai 2023) by mobilising diverse linguistic, semiotic, and sociocultural resources in multilinguals' repertoires to facilitate meaning making (Li 2018). Even though some studies have shown that the integration of translanguaging techniques in English L2 classrooms may boost learners' future L2 practices and promote their interest in English (e.g., Yasar Yuzlu and Dikilitas 2022), less research attention has been paid to exploring the interaction between students' adoption of translanguaging and their LLM in EMI contexts. Given the complex dynamic nature of LLM (Ushioda 2019), the interplay

between translanguaging practices and LLM requires further examination.

This study conducts fine-grained classroom analysis in order to investigate how an EMI teacher's translanguaging practices can contribute to students' LLM. The study will also analyse teachers' reflections on their classroom practices and students' reflections regarding their teachers' translanguaging practices in influencing their LLM. Specifically, the study draws on the Douglas Fir Group's transdisciplinary framework to reconceptualise the framework of L2 motivational self-system (L2MSS) and offer theoretical insights into the role of translanguaging as an important mediator for students' LLM during the transition to EMI contexts.

2 | L2 Learning Motivation and EMI

Motivation can be used to explain why people select a particular activity, how long they are willing to persist at it, and what effort they invest in it (Dörnyei 2001). Extensive research has demonstrated the importance of LLM, with recent studies emphasising situational factors, such as teachers, classroom environment, and activities, that influence students' motivation (Ushioda 2019). This perspective rejects the traditional notion of LLM as an individualised, cognitive and static construct, viewing it instead as a dynamic construct that emerges from the interaction between L2 students and their fluid contexts.

To capture the dynamic and multifaceted nature of LLM, this study draws on the L2MSS framework, which considers the individual and the role of sociocultural contexts in shaping students' LLM. Dörnyei (2009) posits two self-guides: (1) the "ideal L2 self," meaning learners' aspirations of how they will use the L2 in the future; and (2) the "ought-to L2 self," relating to learners' obligations or desire to meet external expectations and avoid undesirable consequences. The third component of the L2MSS is the "L2 learning experience," which bridges learners' self-guides and the situated social learning context and hence will affect levels of motivation. The L2 learning experience includes multiple elements, such as achievements and setbacks, relationships with teachers, the curriculum, and communication with classmates within the L2 learning context (Dörnyei 2009).

Of the three constructs, the ideal L2 self has attracted considerable interest, with many studies showing that having a strong ideal L2 self is a highly effective motivator (e.g., Henry 2009; Al-Hoorie 2018) and is positively associated with improved L2 proficiency (e.g., You and Chan 2015). In contrast, the L2 learning experience has been less studied within this framework. Empirical evidence indicates that students' attitudes toward their L2 learning experience are strong predictors of their motivation to learn the L2 (e.g., Lamb 2012; Papi and Teimouri 2014; Taguchi et al. 2009).

Dörnyei (2019) expanded the definition of the L2 learning experience by conceptualising it as student engagement, which encompasses students' active participation and involvement in school-related activities. These activities may include school contexts, syllabus content, teaching materials, interactions with peers, and relationships with teachers. Although these three constructs have demonstrated transferability across various contexts, evidence also suggests that their applicability may differ across sociocultural settings. For example, in Chinese heritage cultures, knowledge and learning are often deeply tied to one's sense of self, creating a motivational drive that prioritises collective achievement and familial prestige (Ng 2003). Such cultural contexts are considered to foster extrinsic motivation, predominantly driven by meeting others' expectations and preventing disappointment among one's significant others.

Furthermore, while the L2MSS model has illuminated certain aspects of the relationship between context and student engagement with motivation, the situated and occasion-specific nature of this relationship has not been fully explored (Hennebry-Leung and Gao 2023). This study will specifically focus on EMI learning contexts, particularly when students transition from an L1 medium-of-instruction environment to an L2 EMI learning environment. EMI is focused on the acquisition of content knowledge, but it can contribute to increased LLM: since students must grasp the L2 in order to learn the content, L2 learning is more authentic (Swain and Lapkin 2005). Researchers have found that EMI motivates students to improve their L2 English proficiency (e.g., Hennebry and Gao 2021; Hennebry-Leung et al. 2023; Iwaniec and Wang, 2023). Nevertheless, research on the motivational impact of EMI is limited, and further research is needed to explore the interplay between EMI and LLM.

Additionally, there are growing concerns about the lack of support for L2 students in EMI learning contexts (e.g., Evans 2009; Hennebry-Leung et al. 2023). These students' level of English proficiency is often measured in terms of everyday English (Tai and Li 2024), but understanding the content knowledge taught in EMI classes requires the acquisition of subject-specific registers. Having had no experience with instructional content in English previously, HK non-native English speaking students are ill-prepared for learning content subjects in an EMI environment, and they encounter more language challenges in classroom interactions, content understanding, and assignments (Tsou and Kao 2017). Their motivation for English learning also fluctuates (Chen and Kraklow 2015), making adaptation difficult. This explains why motivation, which is widely recognised as a key element in successful L2 learning (Lamb 2017), plays a crucial role in EMI settings. However, there is a lack of research on how L2 learners' adaptation to an EMI environment influences their L2 English learning motivation (e.g., Hennebry-Leung et al. 2023). Examining the relationship between EMI and LLM is crucial for identifying measures to support students' L2 English and content acquisition during their transition to EMI.

3 | L2 Learning Motivation and Translanguaging in EMI Context

To address potential difficulties in maintaining students' motivation throughout their studies and in generating LLM in

EMI classrooms, researchers (e.g., Baker and Fang 2021; Yasar Yuzlu and Dikilitas 2022) suggest using translanguaging as a resource to facilitate students' content and language learning. Translanguaging is a notion that highlights the language users' capacity to utilise their full linguistic and semiotic repertoires to create meaning. As a practical theory of language and an everyday practice, translanguaging recognises that "human communication entails the coordination and interpretation of a vast array of semiotic resources that are entangled with language in fluid and unpredictable ways" (Hawkins 2018, 55–56). Highlighting the transformative capacity of translanguaging, Li (2011) proposes the concept of a "translanguaging space," or interactional space for speakers to mobilise multiple resources to form new meanings. Examples of translanguaging practices in a multilingual classroom to foster students' LLM can include employing multilingual signs, translating L2 academic vocabulary and incorporating bilingual glosses in written texts.

Recent research has examined how translanguaging positively influences students' motivation in L2 learning contexts (e.g., Baker and Fang 2021; Lin and Leung 2024; Yasar Yuzlu and Dikilitas 2022). These studies indicate that when L2 students utilise their diverse linguistic and semiotic resources in L2 learning environments, they can actively engage and exercise their agency during classroom interactions. However, in EMI courses, L2 learning can be demotivating for some students who are reluctant to use the L2 and struggle to understand content through the L2 (Cenoz et al. 2014). To date, there are a considerable number of studies exploring the role of translanguaging in enhancing students' motivation in learning content subjects in EMI settings (e.g., Gu et al. 2024; Jiang et al. 2019; Tai and Li 2021). Nevertheless, while these studies examine students' learning motivation and engagement in learning content subjects in EMI settings, there is a notable lack of research on how translanguaging can support students' LLM in such environments. Thus, further research is needed to investigate the complex and dynamic interaction between students' translanguaging learning experience and LLM. This study specifically examines L2 English learning motivation within the EMI context, recognising that students' engagement with academic English is fundamental to their success in content-based learning. While EMI classrooms aim to facilitate subject knowledge acquisition, the linguistic demands of learning through L2 English create a dual challenge: students must simultaneously develop disciplinary literacy and L2 academic English proficiency. In this paper, LLM is conceptualised as students' drive to master the English necessary to access, process, and articulate subject content, motivated not only by ideal L2 self and ought-to L2 self, but also by the transformative potential of translanguaging pedagogy.

4 | A Reconceptualised Framework of L2 Motivational Self-System

To gain a deeper understanding of how translanguaging is practised and how it interacts with L2 self-guides to influence students' L2 motivational changes, the current research will draw on the notion of translanguaging and Douglas Fir Group's (DFG) transdisciplinary framework in order to reconceptualise

Dörnyei's (2009) L2MSS. The DFG delineates three interrelated levels of influence in L2 acquisition:

- Micro-interactional context: This refers to the moment-to-moment social interactions and the various semiotic tools involved, including language, interactional cues, non-verbal signals, and visual elements.
- Meso-institutional context: This relates to the distinct “small culture” present within schools and classrooms, which is shaped by educational policies, program objectives and design, as well as the dynamics of agency and power during classroom interactions.
- Macro-political/ideological context: This involves broader societal language discourses, policies, prevailing ideologies, prejudices, belief systems, and values.

DFG highlights the importance of recognising how these contextual layers intersect, even when research focuses on a single dimension. Their framework adopts an ecological lens, illustrating how translanguaging practices in EMI classrooms are shaped not only by the meso-level cultures established by educational leaders and teachers, but also by macro-level language policies, ideologies, and wider sociocultural factors (Mendoza et al. 2024; Tai et al. 2025).

However, DFG's framework does not adequately address how LLM operates across these interconnected levels. Given that LLM is influenced by multiple levels of social domains—namely the macro, meso, and micro contexts—and that translanguaging can play a role in shaping students' LLM and identities (Hennebry-Leung and Gao 2023), it is essential to examine students' LLM and their experiences with translanguaging practices in EMI contexts through a multi-layered perspective (Mendoza et al. 2024). To bridge this gap, this paper proposes a contextual framework that aims to draw on DFG's (2016) transdisciplinary framework to guide the study of the role of translanguaging in enhancing students' LLM (Mendoza et al. 2024). Specifically, this contextual framework will enable us to better understand how EMI students' translanguaging learning experiences influence their L2 English motivation in EMI contexts. This framework places LLM at its core, incorporating L2 self-guides (ideal and ought-to L2 selves) as essential components of students' motivation. Additionally, students' L2 self-guides are influenced by their translanguaging learning experience, which reflects various contextual factors at micro-, meso-, and macro-levels.

In this paper, the term “translanguaging learning experience” refers to the translanguaging practices that students engage in within EMI classroom settings and beyond, to aid their content and language learning. The concept of “L2 learning experience” traditionally pertains to the motivational impact of the learning environment and experience (Dörnyei 2009). However, it is important to redefine “L2 learning experience” as “translanguaging learning experience” since language use is context-specific and integrated into the lifeworld, utilising various objects and resources that enable individuals to participate competently in interactions using the L2 (Greer and Wagner 2023). Both neuropsychological and sociolinguistic research have demonstrated that language is intrinsically linked with other cognitive and semiotic systems in the process of meaning and

sense-making (e.g., Cook and Li 2016; Li 2018). Given that L2 learning necessitates contextual practice, a great deal of learning occurs not only in the classroom, but also in various other environments, such as the community and at home, where translanguaging is a common practice (Li 2022). Thus, L2 learning is a translanguaging learning experience, involving students in the use of different linguistic structures, systems, and modalities to construct new knowledge (Zhu et al. 2020).

The reconceptualized framework acknowledges that EMI students' translanguaging learning experience can be shaped in three dimensions: (1) *micro-level*, which highlights that students' translanguaging learning experience can be shaped by their use of translanguaging in or outside the classroom, as well as the teacher's use of translanguaging during classroom interactions; (2) *meso-level*, which emphasizes that students' experiences are influenced by the school's “small culture,” such as the school's language policy; and (3) *macro-level*, which entails broader sociocultural factors, including government policies, beliefs and ideologies, that shape students' translanguaging learning experience (see Figure 1).

By examining these interconnected layers, this framework seeks to offer an ecological perspective on students' L2 motivational development, emphasising the dynamic interplay between students' ideal L2 self, ought-to L2 self, and their translanguaging learning experiences to deepen understanding of students' LLM in EMI contexts. It will also offer a comprehensive understanding of LLM, allowing EMI teachers to pinpoint the specific issues that need to be addressed and the levels at which these issues occur, in their efforts to improve students' LLM.

While the DFG's layered framework has been critiqued for its hierarchical conceptualisation of context (Hult 2019), this study adopts the DFG's model not as a rigid, unidirectional hierarchy, but as a heuristic tool to systematically trace how translanguaging practices and LLM dynamically interact across scales. The tripartite structure (micro-level, meso-level, and macro-level) is retained for its analytical clarity in disentangling complex, overlapping influences, but with critical adaptations to foreground fluidity. First, the reconceptualized framework explicitly rejects linear causality, instead conceptualising translanguaging learning experiences as mediating between levels: macro-level ideologies (e.g., societal monolingual norms) are negotiated at the meso-level (e.g., school policies) and enacted, resisted, or reimagined in micro-level classroom interactions. Second, translanguaging as a practice inherently destabilises boundaries between languages, modalities, and contexts (Li 2018), which introduces dynamism into the model. By centering translanguaging, the reconceptualised framework aligns with Hult's (2019) call for transdisciplinary perspectives that bridge cognitive and socio-cultural ontologies: students' L2 self-guides are continuously reshaped through their agentic engagement with translanguaging practices across contexts. Finally, the framework's emphasis on intersectionality (e.g., how micro-level teacher-student interactions reflect macro-level power asymmetries) addresses critiques of the DFG's “onion” as overly compartmentalised. Hence, although the layered model risks oversimplification, its structured yet flexible design facilitates a systematic analysis of LLM's situatedness across scales—a necessary compromise for empirical research—while translanguaging theory introduces the

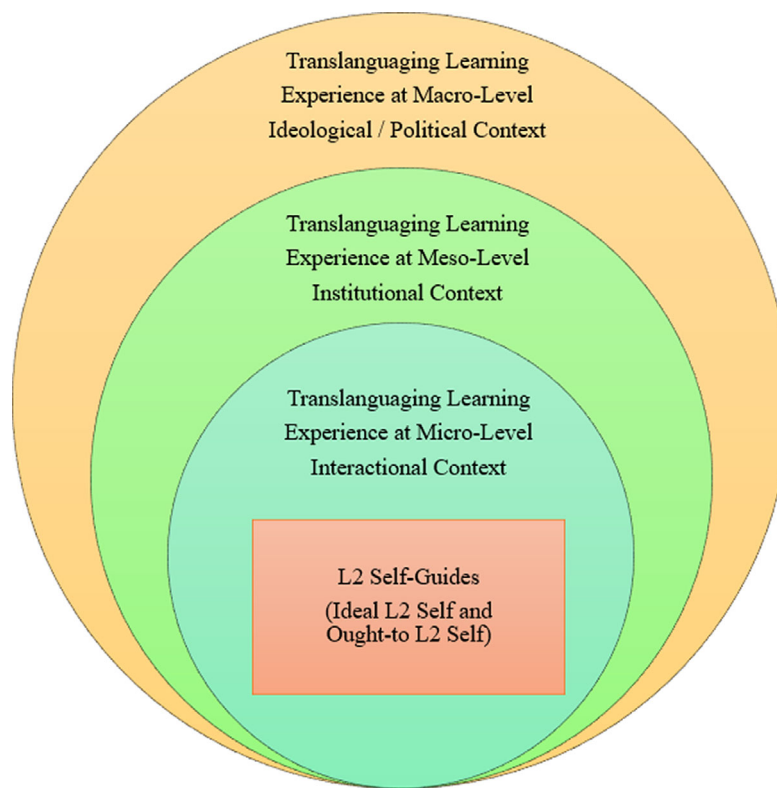


FIGURE 1 | A reconceptualised framework for researching translanguaging and LLM in context. [Color figure can be viewed at wileyonlinelibrary.com]

fluidity needed to prevent the context from being perceived as static or unidirectional.

5 | Methodology

This study adopted a mixed-method approach. A questionnaire was used to assess students' motivation levels in learning L2 English in an EMI context and identify key student participants for follow-up semi-structured interviews and video-stimulated-recall-interviews. I also conducted a detailed classroom analysis in order to investigate how EMI teachers' translanguaging practices can contribute to students' LLM. Finally, I analysed the teacher's reflections on her classroom practices and students' reflections regarding their teachers' translanguaging practices in influencing their LLM. Specifically, this study was guided by the following questions:

1. How does the EMI teacher's creation of translanguaging spaces in EMI classroom interactions shape students' translanguaging learning experience and contribute to students' L2 learning motivation?
2. How do EMI teachers and students make sense of translanguaging practices in EMI classrooms?

5.1 | Participating School

The study was conducted at an EMI secondary school in Hong Kong, providing education from Year 7 to Year 12, following the HK Education Bureau curriculum guidelines. Most classes and exams are in English, except for subjects like Chinese Litera-

ture and Mandarin. While the school's official policy promotes a monolingual EMI approach, in practice, both teachers and students use Cantonese when necessary, especially in Year 7, to ensure comprehension.

5.2 | Participating Teacher

The teacher in the study has one year of experience teaching mathematics in English and is interested in translanguaging to improve student motivation in math and L2 English. She is a native Cantonese speaker, proficient in English and Mandarin, with a background in EMI education and a bachelor's degree in business administration and accounting.

5.3 | Participating Students

In the 3-month classroom observation, I observed a Year 7 class consisting of 34 students. All students in the class had received at least six years of primary education, where Cantonese was the medium-of-instruction at their primary schools. The first language for all students was Cantonese.

To assess students' motivation levels in learning L2 English, a questionnaire was administered. This questionnaire included scales evaluating the constructs of ideal L2 self, ought-to L2 self, and L2 learning experiences in an EMI context. These scales were based on prior research, notably the works of Lamb (2012), Henneby and Gao (2021), Papi and Teimouri (2014), and Taguchi et al. (2009). Cronbach's alpha for the various scales ranged

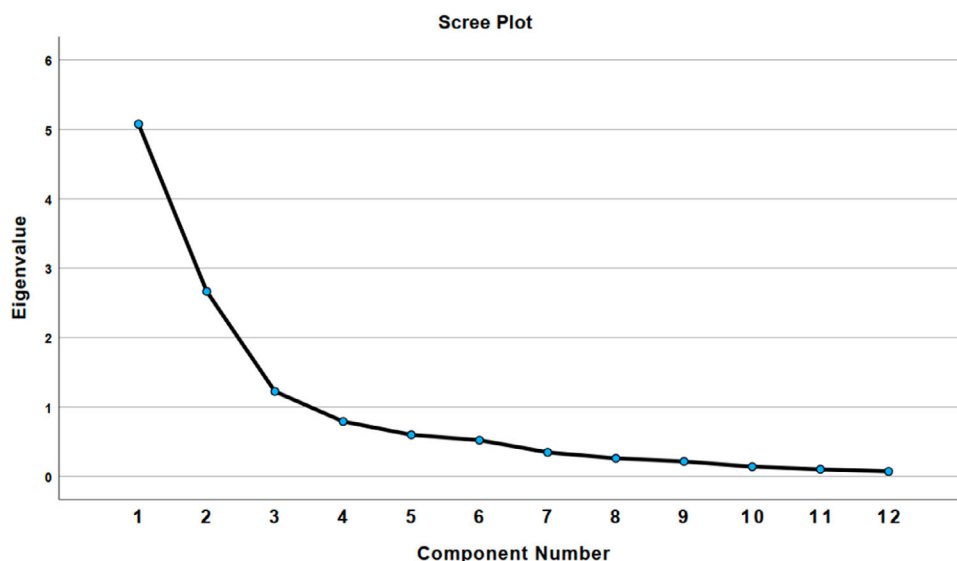


FIGURE 2 | Scree plot. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 1 | Reliability scores for the questionnaire scales.

Construct	Number of items	Reliability (Cronbach's alpha)
Ideal L2 self (ILS)	3	0.81
Ought-to L2 self (OLS)	4	0.80
L2 learning experience in EMI context (LLEEC)	5	0.93

TABLE 2 | Validity of student questionnaire.

KMO		0.746
Bartlett's test of sphericity	Approx. Chi-Square	257.483
	df	66
	Sig.	<0.001

from 0.80 to 0.93, indicating a high level of internal consistency reliability (see Table 1).

Exploratory Factor Analysis supported the suitability of the scale for factor analysis, as evidenced by a Kaiser–Meyer–Olkin value of 0.746, which reflects adequate sampling adequacy (Kaiser 1974). Additionally, Bartlett's test of sphericity reached statistical significance ($p < 0.001$), indicating that the correlation matrix was appropriate for factor analysis (see Table 2). All communalities exceeded 0.4 (Osborne et al. 2008), suggesting strong representation of each item in the factor solution. The scree plot analysis supported a three-factor solution, which explained 75% of the total variance (see Figure 2). While parallel analysis suggested retaining two components (see Table 3), I ultimately adopted a three-component solution based on the following evidence: (1) the scree plot exhibited a clear elbow after the third component (eigenvalue = 1.224), consistent with Cattell's (1966) visual criterion; (2) the third component included theoretically critical items (e.g., Item_5, Item_4, Item_1) that align with the conceptual

TABLE 3 | Parallel analysis results.

Component	Observed eigenvalue	95th percentile random eigenvalue	Decision
1	5.075	2.435	Retain
2	2.664	1.969	Retain
3	1.224	1.643	Do not retain
4	0.790	1.462	Do not retain

framework of L2MSS, and its removal would compromise content validity; and (3) parallel analysis can be overly conservative in small samples ($n = 34$), as noted by Ledesma and Valero-Mora (2007). Though the third eigenvalue fell slightly below the 95th percentile cut-off (1.224 versus 1.643), its proximity to the threshold, combined with the scree plot's inflection point and empirical precedent (Costello and Osborne 2005), supports retaining this component to ensure a substantively meaningful solution. This decision balances statistical rigour with theoretical fidelity. Hence, these results validate the hypothesised three-dimensional structure of the L2 Motivational Self System (see Table 4).

The questionnaire was administered in English and comprised 17 items, each measured on a standard 5-point Likert scale, allowing for thorough analysis while minimising participant fatigue (Dörnyei 2007). It was administered to year 7 students at two intervals: before and after lesson observation, allowing for the identification of students' L2 motivational levels. Based on their scores, students were categorised into high, moderate, and low motivational levels, with the 25th, 50th and 75th quartiles serving as cut-off points for classification. Based on the questionnaire aggregate scores, students with consistent anxiety levels were invited for a video-stimulated-recall-interview.

TABLE 4 | The factor structure of L2 motivational self-system.

	1	2	3
ILS 1	0.67		
ILS 2	0.69		
ILS 3	0.88		
OLS 1		0.83	
OLS 2		0.76	
OLS 3		0.73	
OLS 4		0.82	
LLEEC 1			0.84
LLEEC 2			0.89
LLEEC 3			0.77
LLEEC 4			0.94
LLEEC 5			0.88

5.4 | Data Collection

Before collecting classroom interaction data, the teacher participated in a one-hour professional development session on translanguaging. The researcher introduced the concept of translanguaging and explored strategies for integrating translanguaging into classroom instruction. During the three-month research period, classroom interaction data were collected from the participating teacher. I observed and video-recorded ten one-hour EMI mathematics lessons conducted by the teacher. To gain deeper insights into the teacher's pedagogical practices, a one-hour video-stimulated-recall-interview was conducted. Prior to the interview, video excerpts illustrating significant instances of the teacher's translanguaging practices were selected to serve as stimuli. The teacher was then invited to view these selected video clips and reflect on her pedagogical decisions during specific moments in the EMI lessons. This methodological approach aimed to shed light on how institutional and socio-cultural factors shape EMI teachers' instructional practices (Tai 2023).

Furthermore, a 1 h video-stimulated-recall focus group interview was held with selected students to elicit their perspectives on the effectiveness of translanguaging practices. Based on the questionnaire findings, five students representing all three motivational levels (high, moderate, and low) were invited. Students watched the same video clips as the teacher and discussed their translanguaging experiences in the EMI classroom. During the interview, they were prompted with questions like, "Do you recall this specific classroom interaction?", "How do your teacher's translanguaging practices influence your motivation to learn English?", and "What emotions were you experiencing at that moment?" Students 1, 3, and 10 exhibited low LLM, Student 2 demonstrated high LLM, and Student 5 displayed moderate LLM. Regarding the L2 English proficiency of the focal student participants, the teacher's observations indicated that all students were at the A2 level according to the CEFR scale. However, the students' self-assessments varied: S1 considered her proficiency below average, S2 and S3 viewed their proficiency as average, S4 regarded his skills as low-level, and S5 believed his proficiency was above average.

5.5 | Integrating Multimodal Conversation Analysis With Interpretative Phenomenological Analysis

This study employs a combination of Multimodal Conversation Analysis (CA) and Interpretative Phenomenological Analysis (IPA) to investigate the complexities of translanguaging practices (Tai 2023). Multimodal CA adopts an emic, participant-oriented perspective to analyse how social order is collaboratively constructed through a detailed examination of social interactions. Researchers are encouraged to avoid pre-theorising the relevance of language use, instead focusing on interactional sequences rather than isolated turns or utterances (Hutchby and Wooffitt 1998). Multimodal CA extends traditional CA by integrating multimodal resources, such as gaze, gestures, and object manipulation, viewing these as essential components of social interaction alongside verbal communication. Data were transcribed using Jefferson's (2004) and Mondada's (2018) conventions, with screenshots from video recordings illustrating multimodal interactions in the EMI classroom.

The findings from Multimodal CA are triangulated with video-stimulated recall interview data from both the teacher and students, analysed using IPA. IPA facilitates an understanding of how participants perceive and make sense of their translanguaging practices during specific classroom interactions. This involves a "double hermeneutic" process, wherein researchers interpret how participants themselves interpret their experiences (Smith et al. 2013). IPA thus provides an emic account of participants' lived experiences while also incorporating etic, theory-driven explanations to contextualise psychological phenomena. For clarity, the IPA analysis is presented in a three-column table: the first column features excerpts from the video-stimulated recall interviews, the second captures the teacher's or students' perspectives on translanguaging during those classroom moments, and the third offers the researcher's interpretations, adhering to IPA's two-stage (double hermeneutic) interpretative process.

6 | Analysis

In presenting the findings, I chose to include representative excerpts rather than all transcribed interactions. These excerpts highlight key moments of interaction and exemplify typical uses of diverse multilingual and multimodal resources to support students' L2 English motivation in EMI mathematics classrooms (ten Have 1990). The aim of Multimodal CA analysis is to reveal interactional phenomena within social contexts, rather than simply to confirm the representativeness of selected excerpts (ten Have 1990). Therefore, if the chosen excerpts effectively address the research questions and illustrate the relevant interactional phenomena, their representativeness can be seen as sufficient to ensure the reliability of the findings. During data collection, two cases were identified, showcasing how the teacher uses different multilingual and multimodal resources to construct translingual notes for her students (Extract 1) and address students' lack of motivation towards studying mathematics through L2 English (Extract 2). The analyses of the excerpts are further triangulated by video-stimulated-recall-interviews with the teacher and selected students.

Key Points

H.C.F. (Highest Common Factor)

By prime factorization

e.g. Find the H.C.F. of 54 and 60 by prime factorization.

$$\begin{array}{rcl}
 54 & = & 2 \times 3^3 \\
 60 & = & 2^2 \times 3 \times 5 \\
 \hline
 \text{H.C.F.} & = & 2 \times 3 \\
 & = & 6
 \end{array}$$

To find H.C.F., for each common prime factor, take the one with a smaller index.

FIGURE 3 | The mathematical example.

6.1 | Constructing Translingual Notes for Facilitating Students' Content Learning and LLM

Prior to the extract, the teacher (T) taught students approaches for finding the highest common factor. T projected a mathematical example on the screen (Figure 3), which demonstrates to students that they would need to use prime factorisation in order to find the highest common factor. In order to use prime factorisation, T guides students to first convert the mathematical values into prime factors (lines 15–20).

Extract 1

15 T: product of-?
 16 (1.2)
 17 S3: prime factor
 18 (.)
 19 T: prime factors Δ , 係嘅=
 ((tr. correct))
 --- Δ
 20 T: =將+所有數字+-你嘅 Δ step one Δ 呀!
 ((tr. the first step is, all the numbers))
 +T lifts the pen in her RH to chest level---
 --- Δ

Δ T touches her index fingertips briefly--->Figure #1

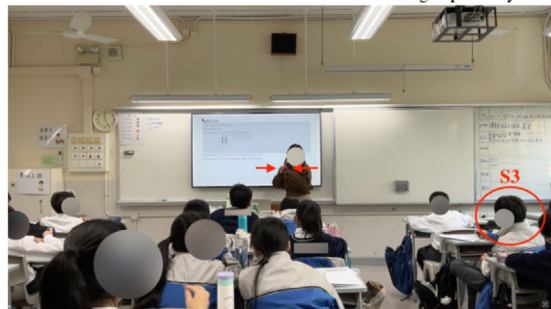


Figure #1

21 + (33.0) +
 +T hunches forward to fetch iPad, holds it at chest level to write “①Express numbers in product of prime factors”, zooms in briefly so Ss can copy it down before zooming out again. T adds “②” under “①”--->Figure #2

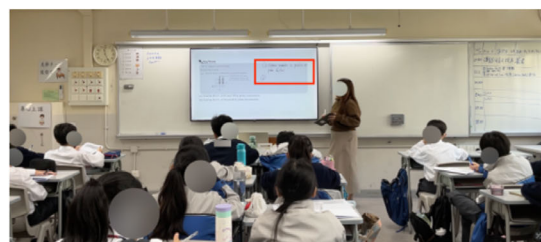


Figure #2

--- Δ
 22 T: +好嘞
 ((tr. alright))
 +T walks towards the right side of the screen--->
 23 (1.1)
 24 T: Δ express 完 Δ
 ((tr. after expressing))
 Δ T taps the blank space next to “②” on the screen with her RH--->Figure #3

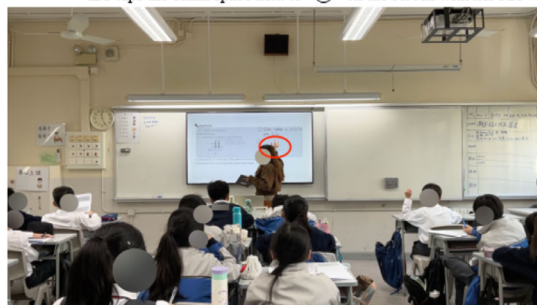


Figure #3

--- Δ
 25 (0.2)
 26 T: 佢as一個product+ of Δ prime factor 嘅 Δ
 ((tr. [the numbers] as a product of prime factors))
 --- Δ
 Δ T turns around, stands next to the breakdown table--->
 --- Δ

27 (0.6)
 28 T: +咁之後嘞
 ((tr. following that))
 +T points at the product of prime factors with her left thumb and index finger--->Figure #4

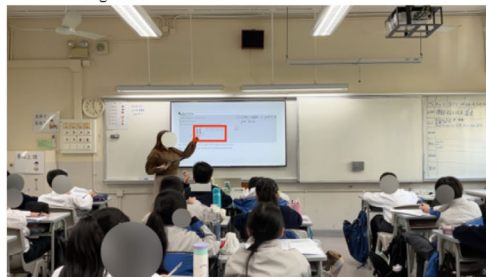


Figure #4

29 (0.2)
 30 T: 佢點樣揀啲數字出嚟呀?
 ((tr. how do we pick the numbers?))
 31 (1.6) +
 --- Δ
 32 T: Δ 佢係點樣去揀嘅? Δ
 ((tr. how do we pick them?))
 Δ T points at the product of prime factors again, with her left index finger--->Figure #5

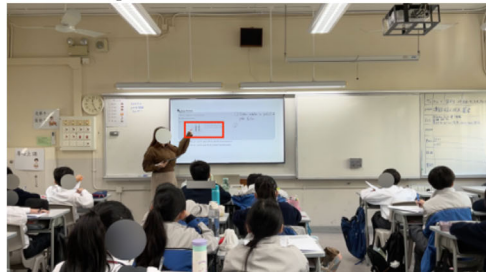


Figure #5

- 33 + (2.6) +
+T lifts her iPad and is about to write, but changes her mind--->
--->+
- 34 T: Δ 佢係要兩邊都有嘅, 定係 Δ =
((tr. do we pick the numbers that are in both rows, or))
 Δ T points at the product of prime factors with her left thumb and index finger--->
---> Δ
- 35 T: =+其中一邊有就係喇?+=
((tr. do we pick those in just one?))
+T circles the bottom row of the product of prime factors with her left index finger--->Figure #6

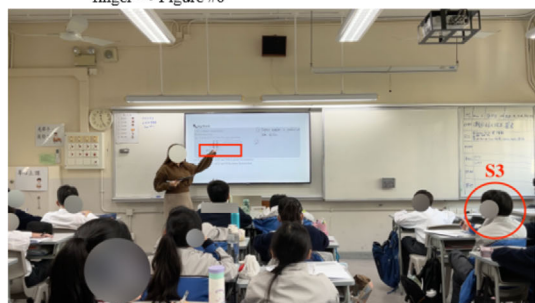


Figure #6

- 36 S?: =兩邊都有=
((tr. both rows))
- 37 S3: =兩邊[都有]
((tr. both rows))
- 38 T: Δ [係喇] Δ
((tr. correct))
 Δ T slightly nods her head--->
---> Δ
- 39 + (0.3)
+T writes “揀2邊都有嘅數字” (“pick the numbers that are in both rows”) next to “②”--->Figure #7

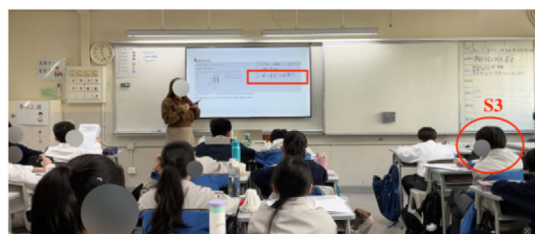


Figure #7

- 40 T: 揀兩邊都有嘅數字
((tr. pick the numbers that are in both rows))
- 41 (18.8) +
--->+
- 42 T: Δ 咁如果佢今次係三個數嘅話 Δ
((tr. so what if there are three numbers))
 Δ T points at the product of prime factors with her left thumb and index finger, with pauses at each of the three rows (the third row being imaginary)--->Figure #8

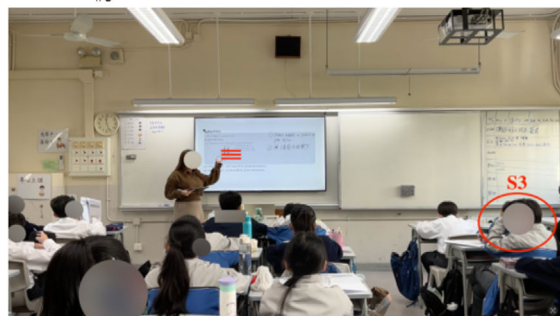


Figure #8

- 43 T: 你要+揀幾多邊呀?
((tr. how many rows do you have to pick from?))
+T hovers her left thumb and index finger over the three rows again--->
- 44 (1.5)
- 45 S3: 三個[數:]
((tr. [if] three numbers))

- 46 T: [揀晒-]
((tr. pick all))
- 47 $\square$ (1.1)
 $\square$ T withholds from pronouncing the word “全” ([chyun4], “all”)--->
- 48 S3: 揀兩邊 $\square$
((tr. pick two rows))
---> $\square$
- 49 (0.3)
- 50 T: 三個數嘅話揀晒+ Δ 全部 Δ 都有嘅數字得唔得?
((tr. pick the numbers that are in all rows if there are three, do you follow?))
--->+
- Δ T flips her LH and does a “wipe” motion twice--->Figure #9

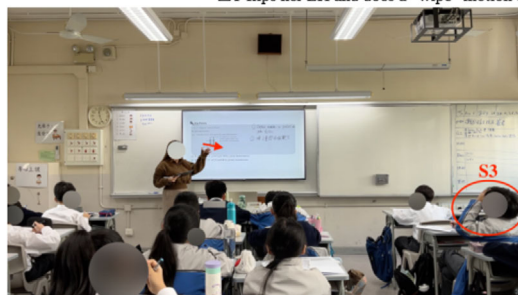


Figure #9

- 51 (0.2)
- 52 T: +咁所以呢個位
((tr. so this part))
+T writes “(全部)” (“(all)”) under “2 邊” (“both rows”)--->Figure #10

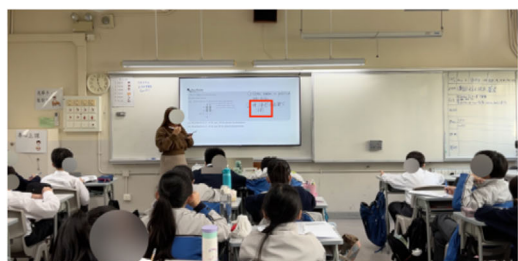


Figure #10

- 53 (1.6)
- 54 T: 全部拉得唔得?+
((tr. all, do you follow?))
--->+
- 55 Δ (4.6) Δ
 Δ T checks the screen while touching her chin and tucking her hair with her RH, takes a step forward--->
---> Δ
- 56 T: +咁呢個次方嘞
((tr. now, the index here))
+T takes a step back, points at the power within the first row of the product of prime factors with her left index finger--->Figure #11

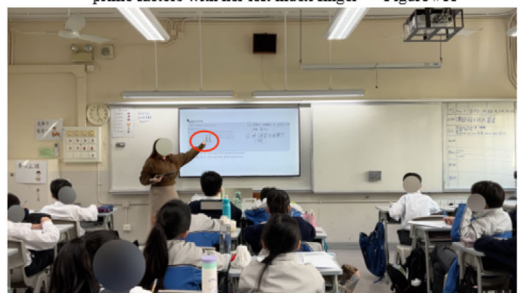


Figure #11

- 57 (0.9) +
--->+

- 58 T: Δ 依度
 ((tr. there is))
 Δ T alternates pointing her left index finger at each of the power in the three rows-->Figure #12

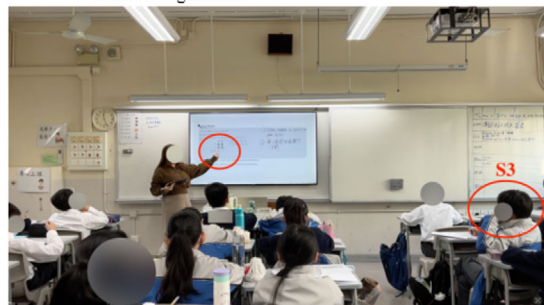


Figure #12

- 59 (0.2)
 60 T: 兩邊都有二嘅=
 ((tr. the number "2" in both rows))
 61 T: =咁個次方寫大嘅定寫細嘅?
 ((tr. so do you write the one with a larger index, or smaller index?))
 62 (1.9)
 63 S3: 寫細
 ((tr. small))
 64 (.) Δ
 ---> Δ
 65 T: $\square$ 寫細嘅口, 係嘞
 ((tr. small, correct))
 $\square$ T nods her head-->
 ---> $\square$
 66 +(6.0)+
 +T writes "③ 次方" ("③index") under "②"-->Figure #13

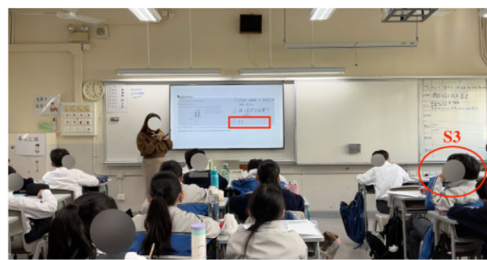


Figure #13

- >+
 67 T: Δ 咁你嘅 index 咁得唔得?
 ((tr. so this is index, do you follow?))
 Δ T adds "(index)寫最細嘅" ("(index) write the smallest one")-->Figure #14

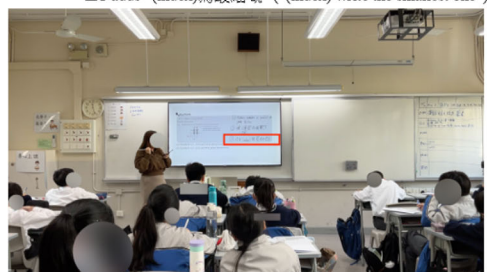


Figure #14

- 68 (11.3) Δ
 ---> Δ
 69 T: o 唔 ok?
 ((tr. not, as in "ok or not?"))
 70 +(0.7)
 +T walks towards the left side of the screen-->
 71 T: 呢個
 ((tr. these))
 72 (.)
 73 T: 係你點樣去搵一個+ Δ hcf 嘅步驟嘅
 ((tr. are the steps you take to find the H.C.F., Highest Common Factor))
 --->+
 Δ T turns around to face the screen-->
 ---> Δ

In line 20, T employs Cantonese to direct students' attention to the initial task step—expressing mathematical values as prime factors. Notably, T switches to L2 English when explicitly stating "step one," emphasising its foundational role in the problem-solving process. During the 33.0-second pause, T writes "① Express numbers in product of prime factors" on her iPad, which is projected on the screen, briefly zooming in to allow students to copy it down (line 21, Figure #2). After completing the writing, T summarises the first step using both Cantonese and English: "express完 (0.2) 佢as一個product+ of prime factor嘞" (lines 24–26). Here, T switches to L2 English for technical terms like "express" and "product of prime factor," potentially highlighting these key concepts to students. Simultaneously, T taps the blank space next to "②" on the screen (line 24, Figure #3), inviting students to consider the next step in finding the highest common factor.

Following a brief pause (0.6s), T initiates the sequence by pointing at the prime factorisation and asking in Cantonese, "佢點樣揀啲數字出嚟" (How do we pick the numbers?) (lines 28–30). Due to the lack of student responses, T then narrows the options by asking, "Do we pick numbers in both rows or just one?" (lines 34–35), accompanied by a circling gesture to indicate the prime factors on the screen (line 35, Figure #6). Students promptly chorus the answer in Cantonese, "兩邊都有" (both rows) (lines 36–37), which T affirms with a nod and a verbal acknowledgement, "係嘞" (correct) (line 38). Finally, T consolidates the answer by writing, "揀2邊都有嘅數字" (pick the numbers in both rows) during a 0.3-second pause (line 39, Figure #7). It is worth noting that T's writing combines written Cantonese ("揀2邊都有嘅") and standardised written Chinese ("數字"), reflecting a blend of colloquial and formal language.

From lines 42–50, T uses Cantonese to invite students to determine how many rows they need to pick from if there are two numbers from each side. In line 45, a student tentatively responds with "three numbers". T then elaborates in line 50, clarifying that if there are three numbers, they should select all numbers present in all rows. In line 52, T writes "全部" (all) under "2邊" (both rows) on her iPad, visually reinforcing the concept of selecting all relevant numbers for finding the highest common factor (Figure #10).

In line 56, T shifts focus to the concept of the index (次方), pointing at the power within the rows of the product of prime factors (Figure #11). Continuing with the use of Cantonese, T guides students to identify the index number "2" in both rows (lines 58 and 60) and poses a critical question in line 61 about whether to write the larger or smaller index. A student responds with "寫細" (write small) in line 63, which T confirms in line 65 with a nod and verbal affirmation, reinforcing the correct answer. Subsequently, T writes "③ 次方(index)寫最細嘅" (index write the smallest one) under point 2, visually structuring the information (lines 66–67, Figures #12 and #14). It is noted that the mathematical term 'index' is first written in standardised Chinese ("次方") and then translated into L2 English as 'index' for students to recognise the English translation. The second part of the notes, "寫最細嘅" (write the smallest one), is written in Cantonese, which potentially enables her students to understand the meaning of the notes.

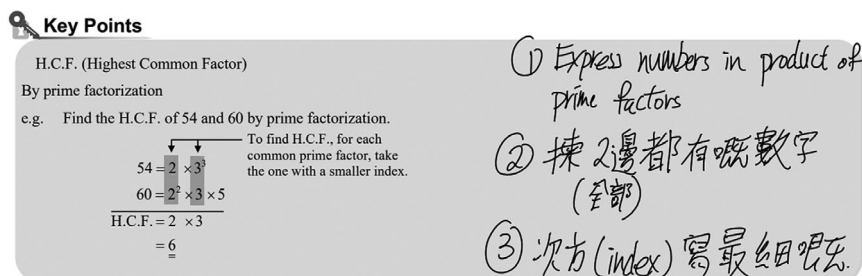


FIGURE 4 | Translingual notes developed by T.

In this extract, T skillfully employs a variety of linguistic and multimodal resources to teach students how to find the highest common factor. T leverages the technological affordances of an iPad, including zooming in and out and projecting her writing on the screen, to enhance her instructional delivery. Notably, T uses a combination of written Cantonese, standardised written Chinese, and English to create translingual notes that help students memorise the step-by-step process of finding the highest common factor. The notes, as shown in Figure 4, encapsulate the comprehensive guidance provided by T throughout the lesson. It is important to note that Cantonese is primarily a spoken language and not traditionally used in written form. T's integration of Cantonese and standardised Chinese in the written notes challenges conventional writing practices and the boundaries of named languages. Additionally, T uses English to indicate key mathematical terms (e.g., “express,” “product of prime factors,” “index”), ensuring that these concepts are clear and accessible to students.

6.2 | EMI Teacher's and Students' Perspectives Towards the Role of Translingual Notes for Facilitating Students' Content Learning and LLM

During the video-stimulated-recall-interview with representative students, they commented on how T's translingual notes sustained their motivation for learning both L2 English and mathematics (Table 5):

In the interview, students report that written Cantonese—their spoken language—enhances comprehension and memory retention of mathematical content. S1, who displays a low-level of LLM, attributes this to simplified cognitive processing, while S5 (moderate LLM) notes that colloquial instructions improve exam recall, directly linking this accessibility to performance outcomes. This suggests that when the meso-level EMI policy does not align with students' everyday language, it can increase cognitive load and make complex concepts less accessible. This also highlights the role of examinations as a macro-level contextual feature in shaping S5's ought-to L2 self. Success in school examinations is important to students and can bring praise and honour, while failure can result in shame, especially in the context of parental expectations in HK (e.g., Chen and Kraklow 2015; Hennebry-Leung and Gao 2023).

Meanwhile, S2 (high LLM) emphasises strategic English use for learning key concepts, framing L2 terms as tools for exam success. This aligns with her “ought-to L2 self,” driven by sociocultural

pressures to memorise L2 English mathematical vocabulary. Both S2 and S5 reveal how translinguaging balances macro-level exam demands with micro-level students' needs: Cantonese reduces cognitive load, while L2 English aids in learning crucial English mathematical terms, thereby promoting extrinsic LLM focused on academic success. From T's perspective, constructing translingual notes that combine written Cantonese, written standardised Chinese and L2 English can facilitate students' mathematical learning and support their LLM towards learning key mathematical terms in English (Table 6):

T's translinguaging practices strategically cultivate students' LLM by aligning linguistic choices with their learning needs. By embedding L2 English mathematical terms (e.g., “products of prime factors”), standardised written Chinese and written Cantonese within the notes, T scaffolds students' exposure to L2 English in a low-anxiety environment. This approach directly supports students' ought-to L2 self, as students recognise the pragmatic value of mastering these terms for exam success in an EMI context (meso-level and macro-level contexts). Crucially, the use of written Cantonese, students' L1, validates their linguistic identity, reducing resistance to mastering L2 English technical terms and fostering LLM by bridging spoken language with formal academic content.

While T's primary goal is to promote mathematical comprehension, her selective integration of L2 English terms, prioritising exam-critical vocabulary, ensures students perceive English as a context-specific tool for academic achievement (lines 2–3). This instrumental focus reinforces pragmatic motivation, as students recognise the immediate utility of L2 mastery. Feedback from students (Table 5) underscores the efficacy of this approach: they report recalling T's translingual notes during exams, which enhances their preparedness and indirectly normalises learning L2 English technical terms as a pathway to success. However, this compromise leads to limited L2 English exposure during classroom interactions—an intentional adjustment to balance institutional requirements (meso-level contextual feature) with students' current language abilities and emotional readiness to engage with L2 English in an EMI environment (micro-level contextual feature).

Hence, T's translinguaging practices operate within a meso-level EMI policy that prioritizes L2 English proficiency, a meso-level institutional focus on exam success, and a micro-level context for catering to students' learning needs by validating L1 Cantonese as a legitimate medium for academic learning. By balancing L1 Cantonese accessibility with targeted L2

TABLE 5 | Video-stimulated-recall-interview with students (Extract 1).

Video stimulated recall interview excerpts	Students' perspectives	Analyst's interpretations of students' perspectives
01 K: 我就想你哋去comment吓Miss呢一個寫notes呢種方法, 運用唔同嘅語言資源, 點樣有機會可以提升到你哋嗰個學習動機啊?學英文, 同埋學數學。有冇idea? ((tr. I just want you all to comment on Miss's method of writing notes, using different language resources. How do you think this could potentially enhance your motivation for learning English and learning Maths. Any ideas?)) 02 S5: 明白更深入啲囉, 我都係覺得 ((tr. To understand more deeply, I think.)) 03 S1: 我覺得因為佢寫口語, 其實即係, 即係方便我哋記啦。同埋當佢上緊堂嘅時候, 即係佢寫, 係快過佢寫書面語嘅。即係可能都係可以慳返少少時間啦。同埋我覺得, 即係有啲人其實佢哋唔係好睇得明。即係如果佢哋睇口語佢哋一定, 讀返出嚟, 佢就一定明個意思, 佢係想表達啲咩。但係可能有時書面語佢哋唔係好明, 即係佢講緊啲乜嘢囉。但口語就應該會, 即係你跟住佢讀返出嚟, 你就會get到佢個意思囉 ((tr. : I think because she writes in colloquial language, it's actually more convenient for us to remember. Also, when she's teaching in class, she writes faster in colloquial language than in formal language. This could save a bit of time. And I think some people don't really understand standardized written language. If they see colloquial language, they will definitely understand the meaning when they read it out loud. But sometimes with formal language, they might not understand what is being said. But with colloquial language, you can read it out and get the meaning.)) 04 K: 係啦, 因為都align返你哋嗰個母語, 我哋大家嘅母語, 就容易啲去記。 ((tr. Yes, because it aligns with your mother tongue, our mother tongue, it's easier to remember.)) 05 S5: 可能你話“express”咁樣啦, 我當好似頭先嗰條數咁樣, 佢話45同埋150, 咁你跟住, 都係如果你記得返佢個notes嘅話呢, 你就話首先要“express”咗, 即係搞掂埋啦, 跟住你就會得到次方啦, 跟住佢如果你話記得晒佢啲notes就話要揀最細, 揀最細同埋要揀兩個數字仲要係相同嘅, 咁你就會可能, 我覺得就會考試容易啲記囉 ((tr. For example, when you say “express,” like with the question earlier, it said 45 and 150. If you remember her notes, you would know that first you need to “express” it, then you get the index. If you remember all her notes, you know you have to choose the smallest, choose the smallest, and then choose two numbers that are the same. It might make it easier to remember for exams.))	Students believe that writing the notes using Cantonese as the spoken language is easier for them to understand the mathematical content since it's more difficult to understand the message if it's written in standardized Chinese. S5 adds that remembering specific instructions or methods conveyed in colloquial language can make it easier to recall steps during exams, which can be crucial for performance.	This suggests that when the language of instruction aligns with the students' everyday speech, it reduces cognitive load and aids comprehension. I point out that using language aligned with the students' mother tongue makes it easier to remember. It is evident that S5 understands the step-by-step procedure for finding the highest common factor. The discussion also highlights the role of examinations in shaping Student 5's “ought-to” self. Success in school examinations is important to students and can bring praise and honour, while failure can result in shame, especially in the context of parental expectations (e.g., Hennebry-Leung and Gao 2023).

(Continues)

TABLE 5 | (Continued)

Video stimulated recall interview excerpts	Students' perspectives	Analyst's interpretations of students' perspectives
06 K: 嗯, 即係幫助你考試去容易啲記, 同埋口語會唔會, 即係覺得, 啊, 好似有把聲同你講緊, 定係點樣? ((tr. Hmm, so it helps you remember for exams. And with colloquial language, does it feel like there's a voice speaking to you, or how does it feel?)) 07 S5: 呀呀 ((tr. yes yes)) 08 S1: 即係你會彈返, 即係你做, 可能做返差唔多題目嘅時候, 你就會彈返呢句嘢, 呀, 你要揀咗, 你要做咗呢個先, 即係你就會諗返起佢個句口語囉。即係你就會記得點樣做 ((tr. It's like it comes back to you. When you're doing a similar question, it will come back to you, like, oh, you need to do this first. You'll recall her colloquial phrases, and you'll remember what to do.)) 09 K: 即係幫助你去記 ((tr. So it helps you to remember.)) 10 S2: 跟住我覺得佢, 即係重點啲啲字, 例如index啲啲字, 佢會用英文嚟寫啦, 即係例如, 你會去考試啲啲題目, 哦我見到呢啲, 好似重點, 啲啲字, 咁我就會記得返啲啲notes啦, 跟住我覺得咁樣會容易啲記囉 ((tr. And I think she writes key words, like "index," in English. For example, when you see those key words in exam questions, oh, I recognize these key words, and I remember the notes. I think this makes it easier to remember.))	S2 mentions that writing key terms in English helps in recognizing and remembering them during exams.	This suggests that a strategic use of English for important concepts can reinforce their significance and aid in facilitating the memorization of key academic vocabulary items in English. Student 2 also highlights the role of examinations as a sociocultural factor in shaping Student 2's "ought-to" self, since she needs to work hard to memorize these L2 English technical terms in order to do well in high-stakes school examinations, which will determine whether she can meet the required standards for getting promotion to Year 8.

scaffolding, T helps students to perceive English as a manageable and context-specific skill that contributes to their academic achievement. This dual focus—leveraging L1 for comprehension and L2 for strategic mastery—demonstrates how translanguaging can simultaneously advance subject-specific understanding and LLM, even within high-stakes exam cultures.

[Correction added on November 13, 2025, after first online publication: T's translanguaging practices operational level has been corrected from 'macro-level' to 'meso-level' in this version.]

6.3 | Addressing Students' Lack of LLM

Prior to the extract, T was preparing students for their first-term examination by reviewing a past exam paper with them. She noted that Section A would consist of simpler questions, often with graphs and tables and shorter English descriptions, while Section B would contain more complex mathematical problems and longer English descriptions. In this extract, T tries to calm the students' nerves as they tackle the questions in Section B (lines 15–18).

Extract 2

15 T: 你見 section b
((tr. when you look at Section B))
16 (.)
17 T: +所有嘅題目呢+=
((tr. all of the questions))
+T zooms out on the iPad (synched with the screen) to show the full length of Section B-->
-->+

18 T: +=全部都係[英文]+
((tr. are all in English))
+T swipes left to show the next page-->Figure #5

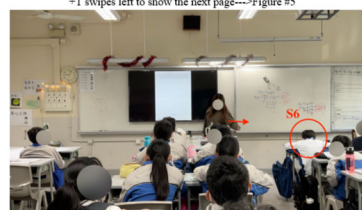


Figure #15

19 S10: [oh no]=
20 T: +=英文+=
((tr. English))
+T swipes left again-->
-->+
21 S7: =囍::=
((tr. whoa))
22 T: +=英文+=
((tr. English))
+T swipes left again-->
-->+

23 Ss: =嘩:::=
 ((tr. whoa))
 24 T: =+[英文]+
 ((tr. English))
 +T swipes left again--->
 --->+
 25 S10: [do not↑] agree↑=
 26 S?: =我想死呀=
 ((tr. kill me now))
 27 T: =+得唔得+
 ((tr. do you follow))
 +T swipes all the way back to the first page of Section B--->
 --->+
 28 (0.3)
 29 S6: y::es
 30 (0.4)
 31 T: +咁所以呢+
 ((tr. so therefore))
 +T zooms in to show the first Section B question more clearly--->
 --->+
 32 (0.5)
 33 T: +△就考吓你去睇嘅△=
 ((tr. [it] tests you))
 +T turns sideways and keeps her eyes on the screen--->
 △T lifts her LH to chest level and turns it multiple times--->Figure #16

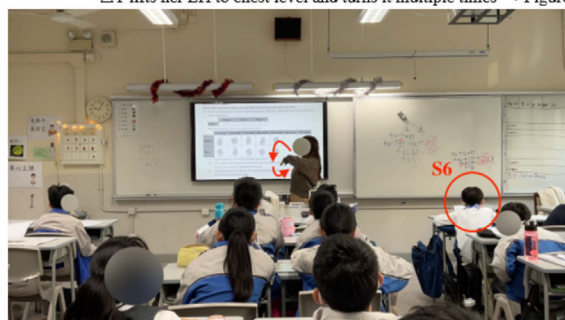


Figure #16

34 T: =□英文嘅□一個能力嘅嘅 hh+ section b=
 ((tr. your ability to read English, this Section B))
 □T rotates both hands, one above the other at chest level--->Figure #17

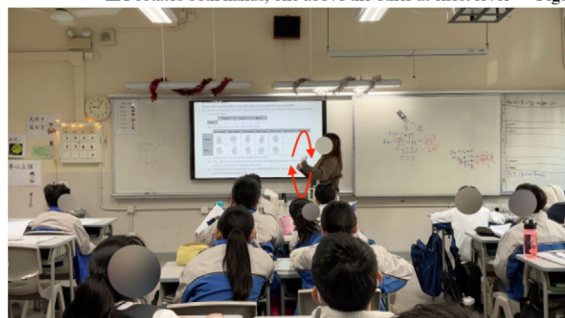


Figure #17

--->□
 --->+
 35 S?: =XX=
 36 S10: =零分=
 ((tr. zero marks))
 37 T: =+裡面<全 hh 部都係文字題啦+
 ((tr. it is all sentence questions))
 +T adjusts the screen slightly--->+
 38 (0.4)
 39 T: +但係仲有一樣嘢你要留意啲咩呀?
 ((tr. but there is one thing you need to pay attention to, what is it?))
 +T lifts RH to touch her hair--->
 40 (0.3)
 41 T: △講起:↑文:字題△喇
 ((tr. when it comes to sentence questions))
 △T lifts LH to chest level as emphasis then drops it--->Figure #18

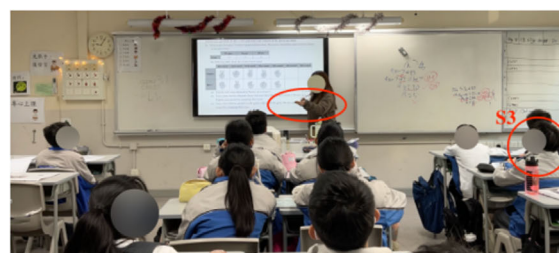


Figure #18

--->△
 42 (.)
 43 T: 我哋最需要留意啲咩呀?
 ((tr. what do we have to pay attention to the most?))
 44 (0.3)
 45 S3: 文字
 ((tr. the text))
 46 (0.2)
 47 T: △答!案嘅話
 ((tr. when it comes to the answer))
 △T pinches LH fingers and gestures forward when saying "answer"--->Figure #19

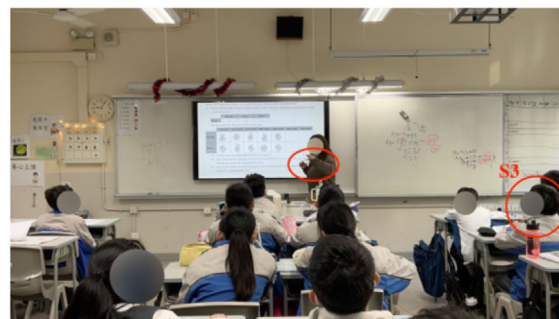


Figure #19

48 (.)
 49 T: 答案嘅話△=
 ((tr. when it comes to the answer))
 --->△
 50 T: =□你需要留意-?
 ((tr. what do you pay attention to?))
 □T unfolds LH, remains palm-up--->Figure #20

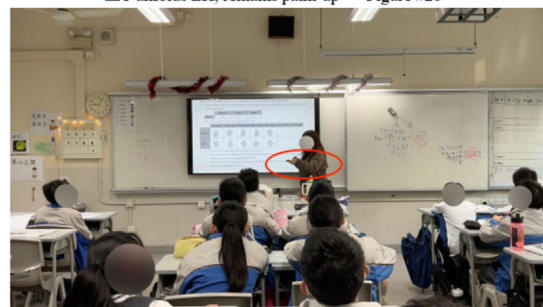


Figure #20

51 (0.2)
 52 T: 仲有咩=-
 ((tr. what else))
 53 Ss: =單位
 ((tr. unit))
 54 (.)
 55 T: △係喇△=
 ((tr. correct))
 △T gestures LH forward as emphasis--->
 --->△
 56 T: =需要留意□+△你要有個單:位:呀=
 ((tr. [you] need to pay attention to the unit))
 --->□
 --->+
 △T does a "chop" sign with both hands multiple times--->Figure #21

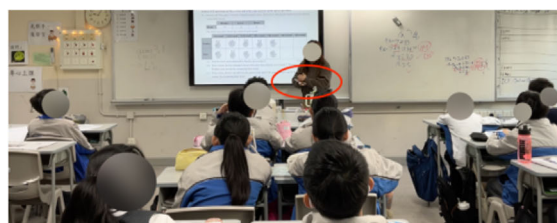


Figure #21

- 57 T: =咁所以你 sectionΔ b
 ((tr. so [in] your Section B))
 --->Δ
- 58 (.)
- 59 T: 你有樣嘢+你好需要留意嘅
 ((tr. you have one thing you must pay attention to))
 +T zooms in on her iPad and starts writing--->
- 60 (0.5)
- 61 T: Δ你嘅 unit 呀↑
 ((tr. it is your unit))
 ΔT adds “☆ Unit” where Section B begins--->Figure #22

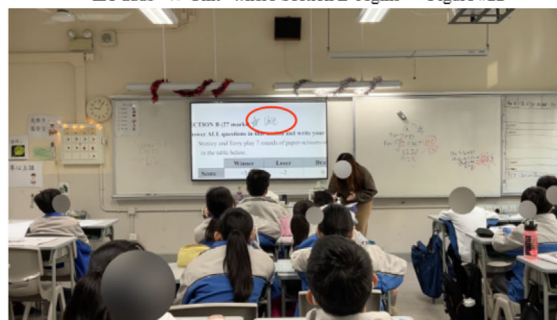


Figure #22

- 62 (0.2)
- 63 T: 得唔得
 ((tr. do you follow))
- 64 (0.6)
- 65 S?: 可唔可以Δ+放棄呀↑
 ((tr. can we give up already))
 --->Δ
 --->+
- 66 □ (0.3)
 □T zooms out on the iPad to show the full question--->
- 67 T: +唔可以+
 ((tr. no [we] cannot))
 +T slightly shakes her head--->
 --->+
- 68 (0.4)
- 69 S?: 痴:線:hh㗎□
 ((tr. [this is] crazy))
 --->□
- 70 (1.9)
- 71 T: 你考試都唔會放棄㗎嘛, +係咪?
 ((tr. you don't give up on your exams, do you?))
 +T looks up to check the screen--->
- 72 (0.2)
- 73 S?: 會=
 ((tr. [we] do))
- 74 S?: 會=
 ((tr. [we] do))
- 75 Ss: 會=
 ((tr. [we] do))
- 76 T: =hahaha+
 --->+
- 77 (.)
- 78 S?: 一見到文字就轉 XX
 ((tr. at the sight of sentence questions, [I] turn))
- 79 (0.4)
- 80 T: Δ唔:好Δ放棄住
 ((tr. do not give up yet))
 ΔT slightly shakes her head--->
 --->Δ

In this extract, T primarily uses Cantonese to advise students on managing their expectations and adopting strategies for handling

long English questions. In lines 15–18, T uses both Cantonese and English to draw students' attention to “section B” (line 15) and indicate that all questions are in English: “你見section b所有嘅題目呢全部都係英文” (when you look at section B, all the questions are in English). Simultaneously, T swipes left on her iPad to show the next page of the exam paper (Figure #15), indicating her intention to continue outlining the paper's structure.

Student 10 (S10), who has a low level of LLM according to the questionnaire findings, exclaims “oh no” in line 19, which potentially underscores his reluctance to engage with the long mathematical questions in English. From lines 20–24, T repeatedly utters ‘英文’ (English) and keeps swiping the examination papers to the left in order to highlight the considerable number of extended questions in English. One student exclaims in Cantonese ‘嘩:::’ (whoa) with extended sounds, indicating his surprise at seeing so many mathematical questions with long English sentences. Another student, in line 26, expresses his desire to die, ‘我想死呀’ (I want to die), exaggerating his/her unwillingness to engage with the examination questions. In lines 31–34, T continues in Cantonese, reminding students that to do well in section B, they need to tap into their L2 English proficiency (Figures #16 and #17), ‘咁所以呢 (0.5) 就考吓你去睇嘅英文嘅一個能力嚟嘅’, highlighting the role of L2 English in achieving good marks in the school examination. In line 36, S10 self-initiates a turn by saying “zero marks” (‘零分’) in Cantonese, further exaggerating his inability to answer the mathematical questions and potentially reflecting his lack of motivation and confidence in achieving good marks.

Although T highlights the difficulty of the school examination, T initiates a new turn to provide advice. In line 37, T points out that section B involves sentence questions, “文字題”, and continues in L1 Cantonese to prompt students to consider what they need to pay attention to in relation to these questions: “但係仲有一樣嘢你要留意啲咩呀 (0.3) 講起:文:字題喇 (.) 我哋最需要留意啲咩呀?” (lines 37–43). In line 45, Student 3 responds with ‘文字’ (words), which is not the preferred answer. From lines 47–52, T continues prompting students to think about what they need to focus on when dealing with sentence questions in the mathematics examination. Several students in line 53 utter ‘單位’ (unit), which T immediately acknowledges as the preferred answer (line 56). To emphasise the importance of considering the mathematical unit, T zooms in on her iPad screen, inviting students to look at the screen, and says “你嘅unit呀” (it's your unit) in line 61. T also uses a symbol of a star sign to add a note for students, “☆ Unit,” next to ‘section B’ (Figure #22) to publicly display this key information.

In line 65, a student expresses a desire to give up, asking, “可唔可以放棄呀” (can we give up already), in line 65. T responds firmly with “唔可以” (no, we cannot) in line 67, while slightly shaking her head to emphasise the refusal. A brief pause follows (line 68), and a student reacts by saying “痴線” ([this is] crazy) in line 69. This expression of frustration is met with a moment of silence (line 70). T then addresses the class, stating, “你考試都唔會放棄 嘛, 係咪?” (you don't give up on your exams, do you?) in line 71. Interestingly, students respond almost immediately and in unison, asserting “會” ([we] do) repeatedly in lines 73–75. T reacts with laughter, “hahaha” in line 76, acknowledging the students' collective reply with amusement. Another student self-initiates

a turn and comments, “一見到文字就轉XX” (at the sight of sentence questions, [I] turn) in line 78, indicating a tendency to avoid sentence questions. T advises the students, “唔好放棄住” (do not give up yet) in line 80, to reinforce her encouragement.

In this extract, T employs a combination of Cantonese and English, along with the affordances of iPad, such as zoom in/out function, symbol of a star sign, and writing notes, to guide students in managing their expectations and strategies for tackling long English questions in their exams. Despite students expressing reluctance and frustration—evident in their exclamations and comments—T responds firmly but empathetically, using humour and encouragement to maintain their motivation.

6.4 | EMI Teacher's and Students' Perspectives Towards the Role of Translanguaging in Sustaining Students' LLM

During the video-stimulated-recall-interview with selected students, students share their feelings regarding T's pedagogical approach in sustaining their LLM (Table 7):

The interview data reveals several key factors that contribute to developing students' LLM. In the classroom interaction, T addresses the challenge posed by L2 English, particularly for students transitioning from Chinese-medium-instruction primary schools. S10's responses highlight the determination among students not to give up on their exam scores, despite initial frustrations (lines 2 and 4). This demonstrates extrinsic motivation rooted in the desire to succeed academically. S10 also notes that the idea of giving up is more rhetorical than real; students are committed to doing their best, reflecting the high value placed on academic achievement. Moreover, S5 reinforces the importance of problem-solving and seeking help when needed (line 7). This attitude represents the ought-to L2 self, where students feel obligated to persevere and use available resources, such as using their L1, to tackle exam questions rather than giving up completely, which would jeopardise their results. This further highlights the role of examinations in shaping students' ought-to L2 selves (macro-level contextual feature) (Chen et al. 2005). Additionally, the “giving up” rhetoric can be seen as a coping mechanism through humour, which serves dual purposes: it helps students to manage stress and simultaneously elicits empathy from T. This interplay between humour and resilience underscores the social and emotional dimensions of the learning environment, where students seek support from their T.

Both S3 and S10 emphasise the importance of an approachable teacher in fostering student initiative and confidence (lines 10 and 12). When students view their teacher as approachable, they are more likely to ask questions and clarify doubts, which is crucial for developing their motivation to learn mathematics through EMI. This dynamic interaction between T and students highlights the context-sensitive nature of LLM, reinforcing the idea that LLM is not static but dynamic and context-dependent (Ushioda 2009). Thus, T's explicit encouragement and approachable demeanour reduce students' anxiety and boost their motivation to use L2 English for learning mathematics, enhancing their willingness to participate in classroom interactions (micro-level contextual feature).

In Table 8, T acknowledges the challenges that Year 7 students face as they transition from primary Chinese-Medium-Instruction schools to an EMI school, where all content subjects are taught in English (lines 2 and 8). Recognising that students often feel overwhelmed, T emphasises the importance of students accepting this reality as a first step. To enhance students' motivation towards learning mathematics, T employs practical pedagogical strategies, such as teaching students to identify keywords in math problems and stressing the importance of units (lines 2 and 6). These methods, reflecting the micro-level contextual features, are designed to help students quickly understand and solve problems, thereby boosting their content learning motivation and their motivation to learn L2 English.

Although T highlights that students' motivation to learn L2 English is largely driven by the necessity to succeed in other subjects taught in English (line 8), I argue that this does not necessarily mean that students' intrinsic motivation to learn L2 English will grow. While T's use of translanguaging in the extract aims to support students in tackling examination pressure, it can be argued that this approach does not necessarily contribute to the development of the students' ideal L2 self, which involves personal aspirations and intrinsic interests. The pressure to perform well in exams can drive students to focus on short-term goals and immediate results, rather than fostering a deeper, intrinsic interest in learning the content subjects and L2 English. Additionally, the meso-level EMI policy adopted at this school restricts students to using only L2 English in their learning and assessments. This restriction can add to the stress and pressure students feel, limiting their ability to fully engage with and understand the subject content (Tai et al. 2025). As a result, the policy may hinder the development of their ideal L2 self, which involves personal aspirations, intrinsic interests, and a genuine engagement with L2 English beyond immediate academic needs. In order to develop students' ideal L2 self in an EMI environment, it is argued that T needs to create opportunities for meaningful engagement with L2 English that go beyond exam preparation, particularly through the facilitation enabled in the micro-level translanguaging space.

7 | Discussion

This study aims to utilise the reconceptualised framework of L2MSS to support the argument that creating a translanguaging space in EMI classrooms can aid students' LLM and facilitate their transition to a new EMI environment. From a micro-level perspective, the study begins with a detailed analysis of interactions within the EMI classroom (Research Question 1). In Extract 1, the teacher effectively combines Cantonese, standardised Chinese, and English to create translingual notes that assist students in memorising the steps for finding the highest common factor. This strategic use of linguistic resources not only clarifies key mathematical concepts but also boosts students' motivation to learn mathematical terms in L2 English. Conversely, Extract 2 uses a mix of Cantonese and English, along with iPad features such as zooming in/out, star symbols, and note-taking, to help students manage their expectations and strategies for tackling long English questions in exams. T heavily relies on Cantonese to emphasise the importance of L2 English proficiency while visually highlighting key sections of the exam paper on her

TABLE 6 | Video-stimulated-recall-interview with the teacher (Extract 1).

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
01 K: OK。咁你睇到呢，呢度就教呢個Highest Common Factor, 你喺度俾notes啦，俾學生。然之後我哋都見到係，有中文同英文啦，但interestingly, 嗰個中英文係，係完全係好fluid, 介唔介意講吓點解要咁樣嚟去construct, 去create呢啲notes俾啲學生呢？ ((tr. OK. So, we can see here that you are teaching the Highest Common Factor, and you are giving notes to the students. We see that Cantonese, written Chinese and English are used fluidly. Could you explain why you construct and create these notes for the students in this way?))		
02 T: 係，好。咁首先呢，點解我第一個會用英文呢？咁因為呢，products of prime factors呢樣嘢呢，我覺得佢哋已經唔會記得個中文係咩意思啦。即係你諗吓，佢個中文係，sorry, 質數嘅積，係。用質數嘅積嚟表達，咁佢哋會更加混淆。咁所以其實，相反呢一樣用中文嘅話，倒不如我用英文，即係呢啲咁樣，咁數學嘅terms, 我寧願我用返英文，去令到佢哋，啊，更加容易去理解返到底發生緊啲咩事。咁所以我第一個，我會用返英文咁樣去理解，咁所以，其實我好多時候我寫notes, 我要寫中文定寫英文，我會用返究竟呢一個terms會唔會係一個好數學嘅terms呢？咁所以，咁其實因為你products of prime factors呢一樣嘢係一個好數學嘅terms, 所以我就會用返英文去處理啦。 ((tr. Sure. First of all, why do I use English for certain terms? Because for something like “products of prime factors,” I believe they won’t remember the meaning in Chinese. For example, the Chinese term is “質數的積” (product of prime numbers), which might confuse them even more. So instead of using Chinese for explaining such mathematical terms, I prefer using English to make it easier for them to understand what is happening. Therefore, when I write notes, I consider whether the term is a significant mathematical term. Since “products of prime factors” is a key mathematical term, I use English for it.))	T strategically incorporates English mathematical terms into the notes to ensure that students become familiar with the essential terminology necessary for understanding and performing well in exams. The decision to use English for terms like “products of prime factors” is driven by the belief that students might not remember or understand the equivalent Chinese terms, thus opting for clarity and practicality.	This approach recognizes the importance of EMI education, where mastering English technical terms is crucial for academic success in an EMI context.
03 T: 咁跟住之後啦，咁你見第二三，其實我都會係揀中文去做返一個notes嘅。咁個原因係咩？因為始終我寫notes, 最後其實都係想佢哋返去自己去睇，去睇嘅啫。咁所以啦，其實寫notes最主要佢哋要睇得明嘅話，咁佢哋用返佢哋嘅mother language, 佢哋一定最容易去睇得明嘅。再加上尤其是，第二同埋第三，你見其實有乜話好數學嘅terms, 除咗嗰個次方，嗰個index嗰個，所以我有特登寫返個英文俾佢哋啫。咁但係因為，如果佢話有特別好多數學嘅terms, 即係有一啲好專有嘅名詞啦，咁樣樣嘅話，其實我自己prefer返，啊，寫notes嘅話，我可能都會prefer返，用返中文去寫。咁最緊要佢哋返到去，佢哋要睇得明自己係寫緊啲乜嘢嘢。咁同埋因為，我費事有陣時如果寫咗一大串英文嘅話，佢哋又唔想睇，係啦。咁所以有陣時你用返中文嚟寫返呢啲notes嘅話，就會係我比較prefer返嘅一個方法咁樣。 ((tr. Then, as you can see in the second and third parts, I choose to write the notes in Chinese. The reason is that when students review the notes at home, they need to understand them easily. Using their mother language, Chinese, makes it easier for them to understand. Especially in the second and third parts, there are not many mathematical terms, except for “index,” which I have written in English. If there aren’t many specialized mathematical terms, I prefer to write the notes in Chinese. The most important thing is that when they review the notes, they need to understand what they are reading. Additionally, if I write a lot of English, they may not want to read it. Hence, writing the notes in Chinese is my preferred method.))		

(Continues)

TABLE 6 | (Continued)

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
04 K: 我都留意到嘅就係通常我哋寫中文呢, 都係會書面語。但係呢個好interesting就係, 因為我哋廣東話好少寫出嚟。咁介唔介意講吓點解會想, 搵一啲廣東話嘅字, 例如嘅啊, 哈哈。因為我好, 我自己讀書係有老師咁樣做嘅, 咁所以我就好想聽吓你嘅說法囉。 ((tr. I also noticed that usually, when we write in Chinese, it's in a formal written style. But this is interesting because we rarely write in Cantonese. Could you explain why you decided to include some Cantonese words, like “嘅” (ge)? I never had teachers do this when I was a student, so I am very interested in your reasoning.)) 05 T: 因為呢個就係呢, 其實你睇書又好, 你睇即係其他文章都好, 都係一個好書面語。但係其實, 即係如果以我自己, 我點樣會可以令到佢去, 即睇得明, 去咬文嚼字, 其實我自己個腦入面, 我會自己將佢全部變返做廣東話。咁樣樣我會更加好理解。即係你如果, 即係你諗吓如果我呢個位, 我第二個位, 我寫書面, 選擇兩, 選擇兩邊皆有的數字。咁你諗吓啦, 我睇第一眼, 其實我已經睇到, 我已經覺得自己唔想睇啦, 我仲要自己翻譯啲, 係啦。跟住我仲要自己個腦入面諗, 究竟我其實廣東話會點樣樣去睇呢?咁所以我, 其實我寫notes嘅話, 我會用返一個偏向係廣東話嘅方式, 咁可以解除咗佢哋睇嘅時候仲要自己翻譯返做廣東話咁樣樣。跟住同理, 尤其是, 即係有陣時, 佢哋其實, 佢哋其實唔係太鍾意中文呢一個subject嘅。咁所以, 佢哋唔鍾意中文嘅原因係咩呢?因為我都有了解過, 佢哋覺得太多書面語, 佢哋其實, 佢哋唔鍾意睇書面語呀。佢哋覺得, 啊, 書面語其實又唔係我講嘅嘢, 咁即係我講嘅時候, 我又唔會等於我嘅書面語。咁就算我溫書嘅時候, 我其實我係咪都要好書面語咁樣樣去背返晒所有嘢。咁所以呢個位, 即係我大多數寫notes, 我都會用返一個比較廣東話嘅方式, 去令到你哋, 啊, 咁你哋背嘅時候, 都唔使自己翻譯咁樣樣。咁我會覺得比較方便少少咁樣。 ((tr. The reason is that, whether you are reading a book or other articles, they are all in formal written language. But for me, to understand the content better, I mentally convert everything into Cantonese. This helps me understand better. If I write in formal language, like “選擇兩邊皆有的數字” (choose numbers that appear on both sides), at first glance, I don't want to read it, and I would have to translate it in my mind. So, I write notes in a way that resembles Cantonese to make it easier for them to understand without having to translate it themselves. Moreover, some students don't particularly like the Chinese subject because of the formal language. They feel that formal language does not reflect the way they speak. So, when they study, they don't want to memorize everything in a formal style. Therefore, I write notes in Cantonese to make it easier for them to remember without having to translate.)) 06 K: 恩。咁喺呢一個example我睇到都, 我哋都睇到就係, 即係都會係希望可以提升到學生對數學嗰個學習動機啦。但係你個人嘅意見, 你覺得有冇機會都可以提升到學生嗰個, 英文作為第二語言嗰個學習動機呀? ((tr. I see. So, from this example, we can see that you aim to enhance students' motivation to learn mathematics. In your opinion, do you think this approach can also improve their motivation to learn English as a second language?))	T prefers to write most of the notes in Cantonese, the students' mother tongue, to make the material more accessible and understandable. This method reduces the cognitive load and helps students grasp complex concepts more easily. The use of written Cantonese also mirrors T's own learning experiences, where converting standardized Chinese into Cantonese aided comprehension.	T aims to foster a positive learning experience and prevent students from feeling overwhelmed by formal written language, which they may find difficult to relate to and comprehend. The use of Cantonese in notes also addresses the students' dislike for formal Chinese, making the learning process more engaging and relatable.

(Continues)

TABLE 6 | (Continued)

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
07 T: 呢個呢, 因為始終暫時如果以EMI嚟講, 我對佢哋最, 即係一個最basic嘅要求啦, 因為佢哋宜家慢慢咁transit緊上嚟。我對佢哋最basic嘅要求係佢哋需要知道一啲數學專有名詞嘅英文。即係, 咁所以你見我, 其實point one個個位, 我直接用英文嚟寫, 因為point one呢一個位, 但基本上佢考試一定會記得啲啲express呀, 跟住之後product of prime factors呀, 咁樣樣。咁所以呢啲位, 佢一定會見到嘅, 咁所以呢啲位, 我會自己prefer返用英文來寫啦, 兼, 我希望佢哋自己睇嘅時候, 佢哋都會更加容易去睇得明, 到底發生緊啲乜嘢嘢, 乜嘢嘢事, 咁樣樣。咁所以, 跟住另外, 但係你話如果呢個位會唔會特別大去提升佢哋數學嘅學習動機嘅話, 可能會有, 即係用英文嚟學習嘅話, 可能會有少少保留。因為始終佢哋其實呢個位, 我同佢哋寫notes嘅話, 可能其實都會以廣東話為主嘅。咁所以, 但係, 淨係得可能會偏向係一啲比較數學嘅terms會係英文。咁所以其實我會話滲透嘅位唔係特別多, 但係因為我, 其實我寫notes同佢哋, 俾佢哋最主要原因係, 佢哋返到去佢哋要識得, 要睇得明自己到底做緊啲乜嘢嘢。咁所以, 會唔會話呢個動機, 即係會提高佢哋用英文嚟學習嘅動機, 可能有咁大。因為始終佢哋寫同埋睇嘅時候, 都係用緊廣東話。但係對於數學嘅專有名詞嚟講嘅話, 我覺得佢哋(寫嘅?)學習機會比較大。 ((tr. Since students are transitioning to an EMI context, my basic requirement is for them to know the English mathematical terms. That's why in the first point, I wrote in English because these terms, like "product of prime factors," are essential for exams. They need to be familiar with these terms. However, whether this approach significantly boosts their motivation to learn mathematics through English might be limited. Most of the notes are still in Cantonese, except for the specialized mathematical terms in English. The main goal is for them to understand what they are doing when they review the notes. So, the impact on their motivation to learn through English might not be very significant. But for learning mathematical terms in English, I think it provides a good learning opportunity.))	The strategic use of English for specific terms provides an opportunity for students to learn and become comfortable with English mathematical vocabulary.	This dual-language approach aims to balance the need for academic success in mathematics with gradual exposure to L2 English, potentially increasing students' confidence and motivation to learn English over time. T's awareness of the students' needs and preferences highlights a student-centered approach, aiming to create a supportive learning environment that caters to both linguistic and academic goals. While T's primary goal is to promote mathematical comprehension, her selective integration of L2 English terms, prioritizing exam-critical vocabulary, ensures students perceive English as a context-specific tool for academic achievement. However, the trade-off lies in the limited scope of L2 exposure—a pragmatic concession to balance institutional demands (meso-level contextual feature) with students' current linguistic capacity and affective readiness to engage with L2 English (micro-level contextual feature).
08 K: 係啦係啦, in terms of學個technical terms, 係啦係啦。我有樣嘢想問, 就係呢, 上一次同學生講到呢一個位, 佢哋都有講到話, 我哋考試嘅時候都會諗返起Miss-T用呢啲比較廣東話嘅notes呀, 有把聲好似同佢哋講緊, 點樣去做返呢一個嘢。咁我想問其實, 呢一個係咪, 都係part of你plan咗嘅一個agenda入面, 定係其實你有冇諗過會有呢一個effect出咗嚟? ((tr. Yes, in terms of learning technical terms. I have one more question. In our previous conversation with students, they mentioned that during exams, they recall the Cantonese notes that you provided and they feel like they can hear your voice guiding them. Is this effect something you planned for, or did it happen unexpectedly?))		

(Continues)

TABLE 6 | (Continued)

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
09 T: 都kind of叫做我plan咗嘅, 因為始終其實如果我唔plan嘅話, 我唔需要寫廣東話。即係如果我唔plan嘅話, 我直接同佢寫書面語, 其實呢個會更加比較作為一個得體少少嘅一個老師嘅做法啦。因為始終其實你見, 即係大家都講到話, 其實好少可會用廣東話嚟寫notes, 咁樣樣。但係因為我, 我熟悉呢一班學生, 所以我會知道佢哋到底需要啲乜嘢嘢。咁所以我會知道佢哋其實唔鍾意書面語, 佢哋鍾意嘅係直接佢哋一睇, 佢哋就會知道呢一樣嘢到底係做緊啲乜嘢嘢。咁所以呢個會係我plan咗話, 我一定要寫返廣東話俾佢哋, 我希望係佢哋考試嘅時候, 佢哋都會記得。就算, 即係佢哋都會記得話, 啊, Miss-T, 佢有同我講過話要揀兩邊都有嘅數字, 咁樣樣去處理, instead of佢哋考試嘅時候就, eh, eh, instead of佢哋考試嘅時候就一頭霧水咁望住。咁所以kind of呢一樣嘢我都plan咗, 希望我自己係做得到嘅, 咁樣樣囉。咁即係如果聽返你嘅feedback, 咁我都覺得我自己應該都做得到嘅, 哈哈 ((tr. It's something that I planned for. If I didn't plan it, I wouldn't need to write in Cantonese. Writing in formal language might be considered more proper for a teacher. However, since I know this class well, I understand their needs. They don't like formal language and prefer something they can immediately understand. So, I planned to write in Cantonese, hoping they would remember it during exams. I want them to recall that "Miss-T told us to choose numbers that appear on both sides," rather than being confused during the exam. So, yes, I planned for this, and based on your feedback, it seems to be working.)) 10 K: 係, 我哋interview嘅時候啲同學仔都話呢一種嘅notes對佢哋嚟講係好有幫助嘅。 ((tr. Yes, during my interview with selected students, students said that this type of notes is very helpful for them.))	T acknowledges that her choice to write notes in Cantonese was intentional and planned. T understands that her students struggle with formal written language and prefer a more conversational style that they can immediately grasp and relate to.	This is a clear example of differentiated instruction tailored to meet the specific needs and preferences of her students. By choosing to write notes in Cantonese, T is prioritizing comprehension and retention over traditional notions of formality and propriety in teaching. This demonstrates T's innovative and empathetic teaching methods, recognizing the balance she strikes between linguistic accessibility and ensuring a thorough understanding of the subject matter. The feedback provided by the students, as mentioned in the video-stimulated-recall-interview, indicates that students find this approach beneficial, which validates T's instructional choices.

iPad. Both extracts show that students' translanguaging learning experiences in the EMI classroom are shaped by the teacher's use of diverse linguistic, semiotic, and technological resources to support their LLM and transition to a new EMI environment.

The meso-level and macro-level contexts are explored through the analysis of video-stimulated-recall-interviews with the teacher and representative students (Research Question 2). At the meso-level, it becomes clear that the EMI policy is a significant institutional feature that motivates the teacher to challenge the traditional monolingual pedagogical approach in order to enable herself and her students to use diverse communicative resources to construct meanings. In the interview, the teacher emphasises the need to use translanguaging to address the varied learning needs and preferences of students. She leverages her pedagogical knowledge and understanding of her students' preferences, such as their preference for reading Cantonese over standardised written Chinese, to create translingual notes aimed at boosting students' motivation to learn L2 English mathematical terms and helping them memorise the specific mathematical knowledge (Table 6). From the students' perspective, these translingual notes are beneficial as they aid in understanding the content (Table 5). The teacher also highlights the challenges faced by

Year 7 students transitioning from Chinese-medium-instruction primary education to EMI secondary education, noting that most students lack strong motivation to learn L2 English (Table 8). This underscores the impact of the EMI policy on students' LLM, as they are expected to learn some or all content subjects through L2 English, which is also used for assessments (Tables 7 and 8). This motivates the teacher to employ translanguaging practices to create a positive classroom environment, reduce anxiety, and support students' motivation to use L2 English for learning mathematics.

From a macro-level perspective, it is evident that the washback effect of examinations significantly influences students' ought-to L2 self. During the video-stimulated-recall-interviews, students highlighted the role of exams as a sociocultural pressure, shaping students' ought-to L2 selves, as they must work hard to memorise these L2 English technical terms to succeed in school exams, which determine their eligibility for promotion to Year 8 (Table 5). Additionally, when students express a desire to give up on the mathematics exam in Extract 2, this sentiment is more rhetorical than genuine. Students are committed to doing their best, reflecting the high value placed on academic success, while also eliciting empathy from the teacher (Table 6). It can be argued

TABLE 7 | Video-stimulated-recall-interview with students (Extract 2).

Video stimulated recall interview excerpts	Students' perspectives	Analyst's interpretations of students' perspectives
01 K: 喺呢個example我覺得好interesting就係, T係好直接同你哋講, 就係話考試就係咁多英文啦。即係英文係對好多中一學生嚟講係一個challenge嚟嘅, 尤其是好多我哋嘅學生都係喺中文小學過嚟, 咁喺呢一個example就見到T就好explicitly同你哋講, 就唔好放棄。即係keep up with the motivation, 咁樣。我就想聽吓你哋嘅意見, 就係T嗰種嘅方法去鼓勵你哋唔好放棄, 你有啲咩心得? ((tr. In this example, what I find very interesting is that T told you directly that the exam will have a lot of English. For many first-year secondary students, English is a challenge, especially since many of our students come from Chinese-medium-instruction primary schools. So in this example, you can see that T explicitly told you not to give up. That is, keep up the motivation. I would like to hear your thoughts on T's method of encouraging you not to give up. What are your thoughts?)) 02 S10: 其實我覺得我哋片入面話, 哎呀, 放棄啦, 但係其實我想講, 邊個真係會畀考試入面, 放棄自己嘅分數呢。所以其實我覺得呢個係 ((tr. Actually, I think in the video we said, "Oh, just give up," but what I want to say is, who would really give up on their exam scores? So, I think this is...)) 03 Ss: 哈哈哈哈哈 ((tr. hahahaha)) 04 S10: 即係可能有啲, 唔係放棄, 而係真係唔識做, 唔識就係唔識 啦, 係咪先。你唔會考試特登拎本書出嚟偷睇 嘛, 你有可能咁樣。咁所以其實我覺得, 雖然我哋話放棄就放棄啫, 不過我哋, 就係, 就係唔會放棄囉。我哋平時同T嘅相處方式就係咁, 即係話, 其實都係一種講吓笑咁樣, 你唔好放棄呀。但其實我哋係唔會放, 佢, 但其實佢係知我哋唔會放棄囉, 只係, 令成個, 成個課堂好笑啲囉。就係咁 ((tr. I mean, maybe some people, it's not that they give up, but they really don't know how to do it. If you don't know, you don't know, right? You wouldn't purposely take out a book to cheat during an exam, that's impossible. So, even though we said to give up, we wouldn't actually do that. The way we usually interact with T is like this, jokingly saying, "Don't give up." But actually, we wouldn't give up. T knows we wouldn't give up, it just makes the whole class more fun. That's it.)) 05 K: 嗯, 嗯。S5你頭先你話會放棄呀? ((tr. Hmm, hmm. S5, you said you would give up earlier?))	S10 points out that the notion of giving up is more rhetorical than actual; students are committed to doing their best, reflecting a deep-seated value placed on academic achievement.	

(Continues)

TABLE 7 | (Continued)

Video stimulated recall interview excerpts	Students' perspectives	Analyst's interpretations of students' perspectives
06 S2: 唔係呀, 講緊第二個同學 ((tr. No, I was talking about another classmate.)) 07 S5: 咁你俾我講啦, 咁我一定唔會放棄啦。咁你話可能, 例如話, 即係你可能例如話, 佢即係教緊你做嗰個問題, 咁可能你例如話, 你如果有啲嘢真係唔明, 咁你咪問老師囉。咁即係又唔好講到話, 真係唔識, 就梗係唔。唔識就梗係一定要試吓撞吓啦, 例如可能例如話你有啲字唔識嘅話, 咁你咪試吓推算吓佢, 用中文來翻譯返, 你個個文字題講緊乜嘢囉, 就係咁 ((tr. Well, let me speak. I definitely wouldn't give up. For example, if T is teaching you how to solve problems and you don't understand something, you can ask the teacher. You shouldn't say you don't know and give up. If you don't know, you should at least try. For example, if you don't know some words, you can try to infer their meaning, translate them back to Chinese, and understand what the problem is about. That's it.)) 08 S2: 我覺得呢即係佢, 有時會講啲笑話啊, 跟住會講好似, 跟着有啲啲我會覺得, 會令到成個課堂嘅氣氛好放鬆啦。跟住佢, 即係平時啲老師就會好strict, 好嚴嘅, 唔講邊個啦。跟住就係會好似直接講, 好似讀書咁樣, 跟住令到好似成個課堂氣氛好嚴謹嘅。跟住可能有啲同學會驚個老師啦。但我覺得佢咁樣做, 會令我哋好似, 會多啲接觸呀。跟住例如話我哋唔識嘅話, 都好似會夠膽問, 我哋唔識啊, 咁樣, 可唔可以explain, 咁樣樣囉 ((tr. I think sometimes T tells jokes, which makes the classroom atmosphere very relaxed. Normally, teachers are very strict and serious, I won't name anyone. They would just teach in a straightforward way, making the classroom atmosphere very tense, and some students might be scared of the teacher. But I think T's way makes us more willing to interact. If we don't understand something, we feel comfortable enough to ask, "I don't understand, can you explain?")) 09 K: 即係T佢個personality, 或者佢個character比較approachable, 令到你哋都會有個學習動機, 係咪? ((tr. So, T's personality or character is more approachable, making you more motivated to learn, right?)) 10 S3: 我覺得咁樣可以令我哋同佢哋, 唔係, 我哋同T嘅, 即係啲啲, 即係如果你唔識嘅話都唔會咁驚, 可能夠膽問囉, 主動問佢囉 ((tr. I think it makes us not afraid to ask T questions if we don't understand. We would be more willing to take the initiative to ask.))	S5 emphasizes the importance of trying and problem-solving rather than giving up. They mention strategies like asking the teacher for help, inferring meanings of unknown words, and translating English words into Chinese to better understand the material. Students appreciate T's encouragement and explicit instructions not to give up, especially in the context of EMI, which is challenging for many of them to use L2 English to learn mathematics. They find T's approach motivating and supportive. Students highlight the importance of interaction with T, noting that T's use of humour and a relaxed approach makes the classroom atmosphere more enjoyable and less intimidating. This comfort level encourages students to engage more actively and ask questions without fear.	This relates to S5's "ought-to" self, as he expresses a determination to employ various strategies, such as using his L1 (Chinese) to translate the English questions, to tackle examination questions rather than giving up completely during an exam, which would jeopardise his results. This highlights the role of examinations in shaping students' "ought-to" L2 selves, potentially reflecting the values of Chinese society, where parents often expect their children to achieve good results in exams as a means of personal fulfillment and to bring praise and recognition to the individuals. T's explicit encouragement and approachable demeanour are effective teaching methods. These methods help in reducing students' anxiety and increasing their motivation and willingness to participate.

(Continues)

TABLE 7 | (Continued)

Video stimulated recall interview excerpts	Students' perspectives	Analyst's interpretations of students' perspectives
11 K: 你會有啲主動, 有啲種confidence會去take個initiative 去問T, 係咪?S10你想講咩呀? ((tr. You would take the initiative and have the confidence to ask T, right? S10, what do you want to say?))		
12 S10: 因為其實咁樣係, 真係會令到課堂輕鬆啲。即係如果我而家數學堂永遠都好嚴嘅老師, 個氣氛會點呢?就會有人講嘢, 好驚, 好驚, 有人講。 ((tr. Because this really makes the classroom more relaxed. If in a math class, the teacher is always very strict, what would the atmosphere be like? No one would talk, everyone would be scared, and no one would speak.))		

that the teacher's use of translanguaging in the extract aims to help students cope with examination pressure. However, the pressure to perform well in exams can lead students to focus on short-term goals and immediate results, further contributing to the students' ought-to L2 self, rather than their ideal L2 self.

The findings of this study add to the existing literature on translanguaging, LLM, and medium-of-instruction policy in several significant ways. Theoretically, the results support the revised conceptual framework of the L2MSS, which enhances the conceptualisation and contextualization of translanguaging and LLM (section 4). The development of this reconceptualised framework addresses the need for a more "robust and holistic model for examining L2 motivation" (Hu et al. 2025), aiming to offer a comprehensive understanding of the L2MSS by integrating the micro-level experiences of L2 students within broader meso-level institutional and macro-level sociocultural contexts (Hennebry-Leung and Gao 2023). It is only through understanding the multiple forces at different layers that we can reach a more fine-grained comprehension of how students' translanguaging learning experiences shape their LLM. Students' LLM is shaped not only from an individual/static perspective, but also evolves in response to, and is co-constructed with, forces at the micro-, meso-, and macro-levels within the DFG framework.

This paper redefines the concept of 'L2 learning experience' by introducing the notion of 'translanguaging learning experience.' The current understanding of the L2 learning experience only acknowledges that L2 learning is about acquiring comprehensive knowledge of a specific named language (Dörnyei 2019). Therefore, the concept of 'translanguaging learning experience' highlights that the L2 learning experience involves not only students' present experiences and their immediate learning environment (Dörnyei 2009), but also the process of students and teachers using different linguistic structures, systems, and modalities to construct new knowledge, which is translanguaging (Zhu et al. 2020).

Methodologically, this study underscores the advantages of employing a combined framework of Multimodal CA and IPA to

investigate EMI students' translanguaging learning experiences during classroom interactions (Tai 2023). Specifically, Multimodal CA facilitates an examination of how EMI teachers' translanguaging practices support students' LLM within classroom interactions. In contrast, IPA enriches our understanding of the relationship between language learning motivation and translanguaging by offering nuanced insights into the subjective experiences and reflections of both teachers and students. Through IPA, researchers can gain access to teachers' reflective insights on their translanguaging practices, revealing how these strategies are perceived to impact students' LLM. Similarly, students' reflections on their teachers' translanguaging practices provide a window into their motivational responses, uncovering personal narratives and emotional resonances that are not easily discernible through observational methods alone. These retrospective insights enrich the analysis by adding depth to the understanding of how translanguaging practices influence students' LLM, highlighting the interplay between cognitive, emotional, and social factors. Previous research on L2MSS has predominantly relied on questionnaires as the sole data source to assess students' L2 motivational development (e.g., Lamb 2012; Papi and Teimouri 2014; Wong 2020; Ye and Hu 2025). However, questionnaires often provide a static view of motivation and may not capture the nuanced and dynamic nature of students' LLM. As noted in section 2, recent researchers (e.g., Du and Jackson 2018; Hu et al. 2025) have emphasised the advantages of using qualitative methods to gain deeper insights into how students' L2 learning experiences influence their LLM (Ushioda 2009). Therefore, combining Multimodal CA with IPA to analyse classroom interaction data and video-stimulated-recall-interview data enables researchers to investigate the link between translanguaging practices in classroom contexts and students' LLM, which cannot be easily explored through quantitative measurements.

In terms of pedagogical implications, this study emphasises the potential benefits of EMI teachers using translanguaging practices to enhance students' LLM. Specifically, the revised conceptual framework of L2MSS will serve as a useful guide for teachers, highlighting the importance of leveraging students'

TABLE 8 | Video-stimulated-recall-interview with the teacher (Extract 2).

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
01 K: 咁你睇到嗰呢條片就, 見到motivation係一個topic啦。介唔介意講吓你當時, 特登去鼓勵呢班學生, 你覺得點解要去鼓勵佢哋, in terms of有冇機會可以鼓勵到佢哋學英文嗰個興趣啦, 同埋學數學嗰個興趣?哈哈 ((tr. So, you can see in this extract that motivation is the topic. Would you mind talking about why you felt the need to encourage these students at that time? Do you think there's a chance to encourage their interest in learning English and Math? Haha.)) 02 T: 咁始終, 上到嚟中學, 佢哋要面對咁多英文, 你其實你頭先都睇到佢哋嘅reaction係, 哇, 唉, 救命呀, 咁樣樣, 係啦。咁所以, 因為呢一樣嘢其實係expected嘅, 因為始終佢哋對於英文其實個motivation唔係特別大呀。咁所以我會預咗佢哋會有呢一個反應嘅。咁所以, 但係, 同時我亦都需要佢哋認清呢一個事實。即係因為其實, 佢哋冇得避免 嘛, 你入咗嚟呢一間學校EMI教嘅話, 你所有嘢你都係要用英文教。咁所以呢一樣嘢, 我首先, 我覺得你, 唔接受呢個事實還唔使接受呢個事實, 但你需要認清楚呢一個事實先。咁所以我第一步一開section B, 我第一樣嘢同佢哋講, 你一定要認清楚咗呢個事實先 啦, 你有得逃避 啦, 咁樣樣, 係啦。咁跟住之後啦, 咁對於佢哋學英文嘅motivation呢, 咁因為你哋都有睇過我其他堂, 咁其實, 尤其是教percentage嗰課, 我成日都同佢哋講, 你記住讀數學, 佢哋係一個reading, 佢哋係英文嘅reading。佢最緊要嘅其實係, 要幫佢哋, 即係要令到佢哋知道, 你有啲咩keywords你要睇到呢, 咁樣樣。咁所以其實開section B嘅話, 其實我最主要嘅, 其實我想教嘅嘢會係, 佢哋要識得睇呢啲咁樣樣嘅keywords。即係instead of你要由頭到尾睇成條問題, 倒不如你直接睇吓有冇啲咩keywords可以令到你哋更加快, 更加容易直接去理解緊, 到底呢一條題目係要做緊啲乜嘢嘢呢?咁所以我覺得, 即係教佢哋睇keywords會係提升返佢哋嘅motivation嘅呢一樣嘢嘅, 我覺得 ((tr. Well, as they enter secondary school, they have to face so much English. You saw their reactions just now, like “Wow, oh my god, help!” Yes. So, this is expected because their motivation for English isn't particularly high. I anticipated this reaction. So, while I need them to recognize this fact, they can't avoid it. Since they are in an EMI school, everything has to be taught in English. So the first step when I opened section B, I told them they need to recognize this fact first. They can't escape it. Then, in terms of their motivation to learn English, as you've seen in my other lessons, especially the one on percentages, I always tell them that learning Math isn't about reading English. The key is to help them identify the keywords they need to look for. So, when I start section B, my main goal is to teach them to recognize these keywords. Instead of reading the entire question from start to finish, they can quickly understand what the question is about by identifying keywords. I believe that teaching them to look for keywords will enhance their motivation.))	T acknowledges that students face significant challenges when transitioning to an EMI school, where all subjects are taught in English. This adjustment can be particularly daunting for students who do not have a strong motivation for learning L2 English. T believes that teaching students to identify keywords in math problems, rather than reading the entire question, can enhance their ability to understand and solve problems more quickly and accurately. This strategy is seen as a way to boost their motivation to complete math problems.	

(Continues)

TABLE 8 | (Continued)

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
03 K: Motivation in terms of咩呀? ((tr. Motivation in terms of what?)) 04 T: Motivation in terms of去完成呢一條題目, 哈哈哈哈哈 ((tr. Motivation in terms of completing the question, hahaha.)) 05 K: 哈哈哈哈哈 ((tr. hahahaha)) 06 T: Sorry, 即係motivation in terms of去完成呢一條題目。因為始終, 佢哋要認清嘅就係, 係呀, 呢條題目係好長, 係好長呀, 但係你有冇需要去睇晒成串, 即係成條題目呢?咁其實講真, 數學嚟講, 同埋我成日都同佢哋講啦, 你一唔一定要睇晒呢?你唔一定嘅, 你睇keywords囉。你睇唔明條題目唔緊要, 你識, 你認得清你嘅keywords, 你知道其實呢條題目大約想你做咩, 咁你, 就算你真係睇唔明啦, 你撞, 你都有機會撞得中。咁所以呢一個位, 係我開section B啦, 兼我教文字題, 我成日會同佢哋講嘅, 睇keywords啦。咁跟住之後, 另外同埋你見到我一開始我其實都好特別提佢地要留意返嘅units呀啲啲啲。因為以數學嚟講, 即係數學, 難聽啲講, 你嘅分, 即係逐個位逐個位執返嚟 嘛, 係咪?但係啦, 你units你一錯就扣一分啦啲。咁所以呢一個位, 我成日上堂都同佢哋講, 文字題units你一定要記得。咁所以其實你見我問, 同, 即係問佢哋文字題你最緊要啲咩, 其實佢哋都答到我一定要係units咁樣樣。咁所以我會話, 開文字題嚟講嘅話, 即係in terms of完成呢條題目嘅motivation, 我會俾話要搵啲keywords啦。但係in terms of一個用英文嚟學數學motivation嘅話, 我會更多prefer on可能你唔好衰一啲你嘅careless呀, 咁樣樣。去睇返你到底你去做呢一條題目, 你有啲咩要留意。其實你做文字題, 你最需要留意啲咩, 或者你每一條文字題你一定會遇到嘅units呢啲位。咁所以我會特別specific返on呢啲位, 咁樣樣去提返佢哋囉 ((tr. Sorry, I mean motivation in terms of completing the question. Because they need to realize that even though the question is long and complicated, they don't need to read the whole thing. In Math, as I always tell them, you don't have to read everything. Just look for the keywords. Even if you don't fully understand the question, recognizing keywords can help you figure out what you need to do, and you might get it right by chance. So, when I start section B and teach word problems, I always emphasize looking for keywords. Additionally, I specifically remind them to pay attention to units, because in Math, you get points for every little thing, but you lose a point if you get the units wrong. So, I always emphasize the importance of units in word problems. When I ask them about the most important thing in word problems, they always say it's the units. So, in terms of motivating them to complete the question, I focus on keywords. But in terms of using English to learn Math, I prefer to focus on avoiding careless mistakes and paying attention to what they need to watch out for in each question. I specifically emphasize these points to help them.))	T also stresses the importance of paying attention to units in math problems, as mistakes in units can lead to significant loss of points.	

(Continues)

TABLE 8 | (Continued)

Video stimulated recall interview excerpts	Teacher's perspectives	Analyst's interpretations of the teacher's perspectives
07 K: 噃, 我就係想講, 就係嗰次同學生做interview嘅時候, 學生就話, 我首先, 如果我要數學做得好, 我英文, 即係我英文做得好, 我數學先會做得好, 咁樣。咁我想問吓你覺得教呢個數學嘅文字題嘅時候, 會唔會對學生嘅英文學習動機, 會唔會有啲咩影響咁樣呀? ((tr. Hmm, I just wanted to mention that when I interviewed the students, they said that if they want to do well in Math, they need to do well in English first. So, I wanted to ask if you think teaching Math word problems affects their motivation to learn English?)) 08 T: 我覺得其實你頭先講得好, 因為有陣時佢哋, 即係佢哋頭先都覆咗你話一定要英文好, 你先至會數學好。咁所以啦, 你點樣樣可以令到數學好?你就一定要英文好, 英文好你可以咁樣樣做?你就係要去提升你嘅英文。咁所以學數, 用英文嚟學數學嘅話, 一定會提升佢哋學英文嘅motivation。因為佢哋知道, 如果你英文唔好嘅話, 即係唔單止講數學, 你其他EMI嘅subject你一定唔會學得好。因為個原因係咩?因為你根本就睇唔明到底究竟發生緊啲咩事, 咁所以呢一個位會係EMI嘅教學, 即係唔同嘅科目你都一定會影響到佢對英文嘅學習動機。因為講真, 你英文又唔好嘅話, 你全部其實kind of你可以say bye bye 啦, 哈哈, 即係咁講。即係呢, 即係你唔好話say bye bye啦, but你英文你學唔好嘅話, 你其他, 你可能甚至連睇你都睇唔明。即係其他科嘅話, 你尤其是history呢啲咁注重英文嘅呢啲科目嘅話, 你更加會連睇都睇唔明。咁所以其實呢啲EMI課程啦, 唔單只數學啦, 一定會對佢哋學英文嘅動機會增加咗。但係, 我唔擔保係興趣會增加咗。因為始終kind of有陣時, 佢哋會覺得自己係being forced要去學英文呀。因為佢哋係為咗要睇明一條題目, 所以佢哋要學英文。咁所以佢哋對學英文嘅動機, 有嘅, 但係佢哋嘅motivation或者佢哋嘅興趣會唔會增加咗, 呢樣嘢我覺得, 我唔敢寫包單話一定會增加咗 ((tr. I think you made a good point. Sometimes they say they need to be good at English to do well in Math. So, to do well in Math, they need to improve their English. Learning Math in English will definitely boost their motivation to learn English because they know that if their English isn't good, they won't do well not just in Math but in other EMI subjects as well. The reason is that if their English proficiency is low, they won't understand what's going on during the lessons. So, EMI teaching in various subjects will impact their motivation to learn English. If their English isn't good, they can say goodbye to everything else, haha. Not literally say goodbye, but if they don't learn English well, they won't understand anything in other subjects, especially those subjects that heavily rely on English like History. So, EMI courses, not just Maths, will definitely increase their motivation to learn English. However, I can't guarantee that their interest in English will increase. Sometimes they feel forced to learn English just to understand the questions. So, while their motivation to learn English increases, I can't say for sure that their interest in English will increase.))	T suggests that by improving students' English skills, students can perform better in maths and other EMI subjects, which in turn can boost their overall L2 English motivation. T is realistic about the fact that while the motivation to learn English might increase out of necessity, this does not necessarily mean that students' interest in English will grow. The motivation might be driven more by the need to succeed in other subjects rather than a genuine interest in the language itself.	This indicates that the students are primarily motivated by their “ought-to self”—the external expectations and pressures to perform well academically in an EMI context. They recognize the practical need to learn English in order to succeed in subjects like Math and History, which are taught in English. T's strategies, such as emphasizing the importance of keywords and units, cater to this sense of obligation by providing practical tools to help students meet these external expectations. However, this approach does not necessarily contribute to the development of the students' “ideal self,” which involves their personal aspirations, intrinsic interests, and long-term goals. The “ideal self” is driven by a genuine passion for learning and self-improvement, which is not the primary focus of the teacher's current methods. While the teacher's strategies are effective in enhancing motivation through necessity, they may not sufficiently inspire a deeper, intrinsic interest in learning English for its own sake.

multilingual, multicultural, and multimodal repertoires to foster their LLM and enhance both content and language learning. Moreover, the analysis of classroom data offers examples for teacher educators to train pre-service and in-service teachers in adopting translanguaging practices to boost students' LLM in EMI classrooms. These practices will provide students with the necessary language support and scaffolding to navigate the challenges of transitioning to an EMI environment, thereby promoting the professional development of EMI teachers. The findings will also support efforts to foster closer collaboration between Hong Kong primary and secondary schools in creating conditions that ensure students' smooth transition to EMI secondary contexts.

8 | Conclusion

This study underscores the critical role of translanguaging practices in fostering LLM and easing students' transition to EMI environments, framed through a reconceptualised framework of L2MSS. By integrating micro-level, meso-level, and macro-level analyses, the reconceptualised framework enables researchers to analyse and reveal how translanguaging learning experiences are shaped by classroom interactions, institutional policies, and sociocultural pressures. Key findings demonstrate that teachers' strategic use of multilingual, semiotic, and technological resources not only clarifies academic content, but also enhances students' motivation to engage with L2 English, particularly in high-stakes contexts like examinations. Theoretically, the study advances the L2MSS by redefining "L2 learning experience" as a dynamic, translanguaging-driven process that bridges individual agency with broader contextual forces. Methodologically, the combination of Multimodal CA and IPA offers a robust approach to capturing both observable classroom practices and the subjective, reflective dimensions of motivation. Pedagogically, the findings advocate for translanguaging as a tool to support EMI teachers in addressing students' diverse needs, reducing anxiety, and fostering collaborative transitions between educational stages. While this study provides valuable insights, several limitations warrant consideration. Firstly, the study's focus on a single EMI mathematics classroom in HK limits the generalizability of its findings to other academic subjects, educational levels, or sociocultural contexts. Second, the study relies heavily on the teacher's perceptions of students' L2 English proficiency levels and students' self-assessments of their own proficiency. To gain a more objective understanding, future research could incorporate standardised English proficiency assessments to evaluate both general and academic English proficiency, exploring how these proficiency levels influence students' LLM. Additionally, future studies could broaden the scope by examining translanguaging practices across multiple EMI subjects and diverse geographical settings. Longitudinal research that tracks students' LLM over time would provide deeper insights into the sustainability and long-term impact of translanguaging practices in supporting EMI students' LLM.

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References

- Al-Hoorie, A. H. 2018. "The L2 Motivational Self System: A Meta-Analysis." *Studies in Second Language Learning and Teaching* 8, no. 4: 721–754. <https://doi.org/10.1111/ijal.12416>.
- Baker, W., and F. Fang (Gabriel). 2021. "‘So Maybe I’m a Global Citizen’: Developing Intercultural Citizenship in English Medium Education." *Language, Culture and Curriculum* 34, no. 1: 1–17. <https://doi.org/10.1080/07908318.2020.1748045>.
- Cattell, R. B. 1966. "The Scree Test for the Number of Factors." *Multivariate Behavioral Research* 1, no. 2: 245–276. https://doi.org/10.1207/s15327906mbr0102_10.
- Cenoz, J., F. Genesee, and D. Gorter. 2014. "Critical Analysis of CLIL: Taking Stock and Looking Forward." *Applied Linguistics* 35, no. 3: 243–262. <https://doi.org/10.1093/applin/amt011>.
- Chen, J. F., C. A. Warden, and H.-T. Chang. 2005. "Motivators That Do Not Motivate: The Case of Chinese EFL Learners and the Influence of Culture on Motivation." *TESOL Quarterly* 39, no. 4: 609. <https://doi.org/10.2307/3588524>.
- Chen, Y.-L. E., and D. Kraklow. 2015. "Taiwanese College Students' Motivation and Engagement for English Learning in the Context of Internationalization at Home: A Comparison of Students in EMI and Non-EMI Programs." *Journal of Studies in International Education* 19, no. 1: 46–64. <https://doi.org/10.1177/1028315314533607>.
- Cook, V., and W. Li, eds. 2016. *The Cambridge Handbook of Linguistic Multi-Competence*. Cambridge University Press. <https://doi.org/10.1017/CBO9781107425965>.
- Costello, A. B., and J. Osborne. 2005. "Best Practices in Exploratory Factor Analysis: Four Recommendations for Getting the Most From Your Analysis." *Practical Assessment, Research, and Evaluation* 10, no. 1: 7. <https://doi.org/10.7275/jyj1-4868>.
- Dörnyei, Z. 2001. "New Themes and Approaches in Second Language Motivation Research." *Annual Review of Applied Linguistics* 21: 43–59. <https://doi.org/10.1017/S0267190501000034>.
- Dörnyei, Z. 2007. *Research Methods in Applied Linguistics: Quantitative, Qualitative, and Mixed Methodologies*. Oxford University Press.
- Dörnyei, Z. 2009. "The L2 Motivational Self System'." In *Motivation, Language Identity and the L2 Self*, edited by Z. Dörnyei and E. Ushioda, 9–42. Multilingual Matters.
- Dörnyei, Z. 2019. "Towards a Better Understanding of the L2 Learning Experience, the Cinderella of the L2 Motivational Self System." *Studies in Second Language Learning and Teaching* 9, no. 1: 19–30. <https://doi.org/10.14746/sslt.2019.9.1.2>.
- Douglas Fir Group. 2016. "A Transdisciplinary Framework for SLA in a Multilingual World." *The Modern Language Journal* 100: 19–47. <https://doi.org/10.1111/modl.12301>.
- Du, X., and J. Jackson. 2018. "From EFL to EMI: The Evolving English Learning Motivation of Mainland Chinese Students in a Hong Kong

- University." *System* 76: 158–169. <https://doi.org/10.1016/j.system.2018.05.011>.
- Education Bureau. 2009. *Fine-Tuning the Medium of Instruction for Secondary Schools*. Education Bureau Circular No. 6/2009. Government Printer.
- Evans, S. 2009. "The Medium of Instruction in Hong Kong Revisited: Policy and Practice in the Reformed Chinese and English Streams." *Research Papers in Education* 24, no. 3: 287–309. <https://doi.org/10.1080/02671520802172461>.
- Greer, T., and J. Wagner. 2023. "The Interactional Ecology of Homestay Experiences: Locating Input Within Participation and Membership." *Second Language Research* 39, no. 1: 85–113. <https://doi.org/10.1177/02676583211058831>.
- Gu, M., G. Jiang, and M. Chiu. 2024. "Translanguaging, Motivation, Learning, and Intercultural Citizenship Among EMI Students: A Structural Equation Modelling Analysis." *International Journal of Intercultural Relations* 100: 101983. <https://doi.org/10.1016/j.ijintrel.2024.101983>.
- Hawkins, M. R. 2018. "Transmodalities and Transnational Encounters: Fostering Critical Cosmopolitan Relations." *Applied Linguistics* 39, no. 1: 55–77. <https://doi.org/10.1093/applin/amx048>.
- Hennebry, M., and X. Gao. 2021. "Interactions Between Medium of Instruction and Language Learning Motivation." *International Journal of Bilingual Education and Bilingualism* 24, no. 7: 976–989. <https://doi.org/10.1080/13670050.2018.1530190>.
- Hennebry-Leung, M., and X. Gao. 2023. *Language Learning Motivation in a Multilingual Chinese Context*. Routledge. <https://doi.org/10.4324/9781003025122>.
- Hennebry-Leung, M., W. T. Tseng, and X. Gao. 2023. "Language Learning Motivation, Learner Agency, and the Medium of Instruction." *Psychology in the Schools* 60: 1–17. <https://doi.org/10.1002/pits.22935>.
- Henry, A. 2009. "Gender Differences in Compulsory School Pupils' L2 Self-Concepts: A Longitudinal Study." *System* 37: 177–193. <https://doi.org/10.1016/j.system.2008.11.003>.
- Hu, X., Y. Zhang, and M. Hennebry-Leung. 2025. "Conceptualising the Actual L2 Self Into the L2 Motivational Self System." *System* 130: 103620. <https://doi.org/10.1016/j.system.2025.103620>.
- Hult, F. M. 2019. "Toward a Unified Theory of Language Development: The Transdisciplinary Nexus of Cognitive and Sociocultural Perspectives on Social Activity." *The Modern Language Journal* 103: 136–144. <https://doi.org/10.1111/modl.12527>.
- Hutchby, I., and R. Wooffitt. 1998. *Conversation Analysis: Principles, Practices and Applications*. Blackwell.
- Iwaniec, J., and W. Wang. 2023. "Motivations to Enrol in EMI Programmes in China: An Exploratory Study." *Applied Linguistics Review* 14, no. 6: 1563–1587. <https://doi.org/10.1515/applirev-2021-0180>.
- Jefferson, G. 2004. "Glossary of Transcript Symbols With an Introduction." In *Conversation Analysis: Studies From the First Generation*, edited by G. Lerner, 14–31. John Benjamins.
- Jiang, L., L. J. Zhang, and S. May. 2019. "Implementing English-Medium Instruction (EMI) in China: Teachers' Practices and Perceptions, and Students' learning Motivation and Needs." *International Journal of Bilingual Education and Bilingualism* 22, no. 2: 107–119. <https://doi.org/10.1080/13670050.2016.1231166>.
- Kaiser, H. F. 1974. "An Index of Factorial Simplicity." *Psychometrika* 39, no. 1: 31–36. <https://doi.org/10.1007/BF02291575>.
- Lamb, M. 2012. "A Self System Perspective on Young Adolescents' Motivation to Learn English in Urban and Rural Settings." *Language Learning* 62, no. 4: 997–1023. <https://doi.org/10.1111/j.1467-9922.2012.00719.x>.
- Lamb, M. 2017. "The Motivational Dimension of Language Teaching." *Language Teaching* 50, no. 3: 301–346. <https://doi.org/10.1017/S0261444817000088>.
- Lasagabaster, D., and A. Doiz. 2017. "A Longitudinal Study on the Impact of CLIL on Affective Factors." *Applied Linguistics* 38, no. 5: 688–712. <https://doi.org/10.1093/applin/amv059>.
- Ledesma, R. D., and P. Valero-Mora. 2007. "Determining the Number of Factors to Retain in EFA: An Easy-to-Use Computer Program for Carrying out Parallel Analysis." *Practical Assessment, Research, and Evaluation* 12, no. 1: 2. <https://doi.org/10.7275/wjnc-nm63>.
- Li, W. 2011. "Moment Analysis and Translanguaging Space: Discursive Construction of Identities by Multilingual Chinese Youth in Britain." *Journal of Pragmatics* 43: 1222–1235. <https://doi.org/10.1016/j.pragma.2010.07.035>.
- Li, W. 2018. "Translanguaging as a Practical Theory of Language." *Applied Linguistics* 39: 9–30. <https://doi.org/10.1093/applin/amx039>.
- Li, W. 2022. "Translanguaging as a Political Stance: Implications for English Language Education." *ELT Journal* 76, no. 2: 172–182. <https://doi.org/10.1093/elt/ccab083>.
- Lin, S., and A. Leung. 2024. "ESL Classroom Interactions in a Translanguaging Space." *Applied Linguistics Review* 15, no. 6: 2397–2425. <https://doi.org/10.1515/applirev-2022-0202>.
- Macaro, E. 2018. *English Medium Instruction*. Oxford University Press.
- Mendoza, A., L. Hamman-Ortiz, Z. Tian, et al. 2024. "Sustaining Critical Approaches to Translanguaging in Education: A Contextual Framework." *TESOL Quarterly* 58, no. 2: 664–692. <https://doi.org/10.1002/tesq.3240>.
- Mondada, L. 2018. "Multiple Temporalities of Language and Body in Interaction: Challenges for Transcribing Multimodality." *Research on Language and Social Interaction* 51, no. 1: 85–106. <https://doi.org/10.1080/08351813.2018.1413878>.
- Ng, C.-H. 2003. "Re-Conceptualizing Achievement Goals from a Cultural Perspective." In *Joint Conference of NZARE & AARE*, 20. AARE.
- Osborne, J. W., A. B. Costello, and J. T. Kellow. 2008. "Best Practices in Exploratory Factor Analysis." In *Best Practices in Quantitative Methods*, edited by J. W. Osborne, 205–213. Sage Publishing.
- Papi, M., and Y. Teimouri. 2012. "Dynamics of Selves and Motivation: A Cross-Sectional Study in the EFL Context of Iran." *International Journal of Applied Linguistics* 22, no. 3: 287–309. <https://doi.org/10.1111/j.1473-4192.2012.00312.x>.
- Papi, M., and Y. Teimouri. 2014. "Language Learner Motivational Types: A Cluster Analysis Study." *Language Learning* 64, no. 3: 493–525. <https://doi.org/10.1111/lang.12065>.
- Poon, A. Y. K. 2013. "Will the New Fine-Tuning Medium-of-Instruction Policy Alleviate the Threats of Dominance of English-Medium Instruction in Hong Kong?" *Current Issues in Language Planning* 14, no. 1: 34–51. <https://doi.org/10.1080/14664208.2013.791223>.
- Smith, J. A., P. Flowers, and M. Larkin. 2013. *Interpretative Phenomenological Analysis: Theory, Method, and Research*. Sage.
- Swain, M., and S. Lapkin. 2005. "The Evolving Sociopolitical Context of Immersion Education in Canada: Some Implications for Program Development." *International Journal of Applied Linguistics* 15, no. 2: 169–186. <https://doi.org/10.1111/j.1473-4192.2005.00086.x>.
- Taguchi, T., M. Magid, and M. Papi. 2009. "The L2 Motivational Self System Amongst Chinese, Japanese, and Iranian Learners of English: A Comparative Study." In *Motivation, Language Identity and the L2 Self*, edited by Z. Dörnyei and E. Ushioda, 66–97. Multilingual Matters.
- Tai, K. W. H. 2022. "Translanguaging as Inclusive Pedagogical Practices in English Medium Instruction Science and Mathematics Classrooms for Linguistically and Culturally Diverse Students." *Research in Science Education* 52, no. 3: 975–1012. <https://doi.org/10.1007/s1165-021-10018-6>.
- Tai, K. W. H. 2023. *Multimodal Conversation Analysis and Interpretative Phenomenological Analysis: A Methodological Framework for Researching Translanguaging in Multilingual Classrooms*. Routledge. <https://doi.org/10.4324/9781003351047>.

- Tai, K. W. H., and W. Li. 2021. "Co-Learning in Hong Kong English Medium Instruction Mathematics Secondary Classrooms: A Translanguaging Perspective." *Language and Education* 35, no. 3: 241–267. <https://doi.org/10.1080/09500782.2020.1837860>.
- Tai, K. W. H., and W. Li. 2024. "The Affordances of iPad for Constructing a Technology-Mediated Space in Hong Kong English Medium Instruction Secondary Classrooms: A Translanguaging View." *Language Teaching Research* 28, no. 4: 1501–1551. <https://doi.org/10.1177/13621688211027851>.
- Tai, K. W. H., W. Li, and E. K. Y. Loh. 2025. "Enhancing Students' Content and Language Development: Implications for Researching Multilingualism in CLIL Classroom Context." *Learning and Instruction* 96: 102083. <https://doi.org/10.1016/j.learninstruc.2025.102083>.
- Tai, K. W. H., and L. Wei. 2020. "Bringing the Outside in: Connecting Students' Out-of-School Knowledge and Experience Through Translanguaging in Hong Kong English Medium Instruction Mathematics Classes." *System* 95: 102364. <https://doi.org/10.1016/j.system.2020.102364>.
- ten Have, P. 1990. "Methodological Issues in Conversation Analysis." *Bulletin of Sociological Methodology/Bulletin De Méthodologie Sociologique* 27, no. 1: 23–51. <https://doi.org/10.1177/075910639002700102>.
- Tsou, W., and S. Kao. 2017. *English as a Medium of Instruction in Higher Education: Implementations and Classroom Practices in Taiwan*. Springer Nature.
- Ushioda, E. 2009. "A Person-in-Context Relational View of Emergent Motivation, Self and Identity." In *Motivation, Language Identity and the L2 Self*, edited by Z. Dörnyei and E. Ushioda, 215–228. Multilingual Matters.
- Ushioda, E. 2019. "Researching L2 Motivation: Past, Present and Future." In *The Palgrave Handbook of Motivation for Language Learning*, edited by M. Lamb, K. Csizér, A. Henry, and S. Ryan, 661–682. Springer.
- Wong, Y. K. 2020. "Effects of Language Proficiency on L2 Motivational Selves: A Study of Young Chinese Language Learners." *System* 88: 102181. <https://doi.org/10.1016/j.system.2019.102181>.
- Yasar Yuzlu, M., and K. Dikilitas. 2022. "Translanguaging in the Development of EFL Learners' Foreign Language Skills in Turkish Context." *Innovation in Language Learning and Teaching* 16, no. 2: 176–190. <https://doi.org/10.1080/17501229.2021.1892698>.
- Ye, X., and G. Hu. 2025. "The Effects of Three Motivational Interventions on EFL Learners' L2MSS: A Longitudinal Perspective." *System* 129: 103584. <https://doi.org/10.1016/j.system.2024.103584>.
- You, C., and L. Chan. 2015. "The Dynamics of L2 Imagery in Future Motivational Self-Guides." In *Motivational Dynamics in Language Learning*, edited by Z. Dörnyei, P. D. MacIntyre, and A. Henry, 405–430. Multilingual Matters.
- Zhu, H., W. Li, and D. Jankowicz-Pytel. 2020. "Whose Karate? Language and Cultural Learning in a Multilingual Karate Club in London." *Applied Linguistics* 41, no. 1: 52–83. <https://doi.org/10.1093/applin/amz014>.